Where Does Heterogeneous Federated Learning Lose Rare Classes? A Stage-by-Stage Comparison with Centralised Training

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Abstract

Generator-based model-heterogeneous federated learning (GeFL-F) lets clients with different architectures share knowledge through a frozen feature extractor, a conditional feature generator trained by federated averaging, and per-architecture heads. It was evaluated on balanced clients only. We ask where, under long-tailed clients, the federated pipeline loses accuracy relative to the *same* pipeline trained centrally on the pooled data, and answer stage by stage, with proofs where the question is exact and measurements where it is not.

The client partition is a pure label shift: every client's class-c data are a random subset of the pooled class-c data. Federation therefore never biases a class-conditional target; it loses accuracy only through the optimiser, through parameters coupled to the client's label prior, and through sample size. Each appears once.

Generator. With Adam's coupled weight decay and flat averaging, clients that lack a class shrink its conditioning row at Adam's step size. A step ratio computed from the class counts alone predicts the outcome before training: across 900 logged rows on four datasets and 10–100 clients, all 352 rows with ratio below 1 collapsed (Spearman 0.94 between the ratio and the final norm), and a prediction fixed before the SVHN runs with 50 / 100 clients came true for all 16 flagged rows. Removing the coupled decay recovers 2–11 points of balanced accuracy, and averaging each row over its holders adds 1.3–1.8 at 50–100 clients.

Last layer. Each head's last layer is fitted to one client's label prior. This is the largest loss, 7–20 points. A class-balanced linear readout computed from secure-aggregated class sums of each architecture's embeddings (ER) is *exactly* the centralised readout of the same bodies, and recovers it. Per-client logit corrections — logit adjustment, FedLC, FedRS, balanced softmax — are exact reparametrisations of the bias for a single-client head; on real data alone they give the same ER accuracy within 0.5 points on every seed and dataset.

Representation. A body trained on one client's data loses 3.0–4.2 points against one trained on the pooled data (MNIST, FashionMNIST, two seeds). Synthetic features close this gap only if they carry their label: we prove that label-independent synthetic data act exactly as a penalty on the spread of the logits, and measure that the gain from the synthetic phase changes sign with the generator's signal fraction (CIFAR-100: -2.2 points for GeFL-F, $+1.8$ for an exact-statistics generator). That generator — conditioned on the exactly aggregated class mean, moment-corrected and herded toward the exact class embedding — raises the signal fraction and brings the bodies to the pooled-body level on MNIST and FashionMNIST.

With the same readout for both, it beats GeFL-F by $+2.0$ to $+3.9$ points on MNIST, FashionMNIST, SVHN, CIFAR-10 and CIFAR-100-LT (seeds 0–2). On new seeds, with every method's settings chosen on validation data and nine baselines including server-side ensemble distillation (FedDF) and a Gaussian generator built from exact statistics, its margin over the strongest one is -0.3 to $+1.9$ points (significant on MNIST and CIFAR-100). On MNIST and FashionMNIST the resulting federated pipeline is within 0–2 points of the centralised two-stage pipeline; the remaining long-tail loss (3.2 / 2.7 points centrally) is present without federation. With 10 clients each architecture has one client, the extractor is small and frozen (CIFAR-100 accuracy is 15–19% for every method), and the centralised comparison covers three of the five datasets.

The mechanism is not specific to generators: in federated matrix factorisation on MovieLens-1M, local Adam with coupled L2 collapses 83–85% of the rare item embeddings and costs accuracy on them. There, centralised Adam with coupled L2 already collapses items seen less often than its second-moment memory (about once per 1,000 steps); federation amplifies it as the number of clients grows, and one step-frequency rule predicts both.

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1. Introduction

Motivation. Federated deployments rarely have one model architecture or balanced labels. Hospitals run different networks on different hardware, and rare conditions appear at only a few sites. Generator-based model-heterogeneous FL (GeFL, and its feature-level variant GeFL-F [3]) handles the first problem: a shared generative model, trained by federated averaging, carries knowledge between architectures that cannot be averaged. Its evaluation used IID clients only.

Question. When labels are long-tailed across clients, how far is the federated solution from the centralised one, and which stage is responsible? We compare every stage of GeFL-F with its centralised counterpart on the same pooled data and the same frozen-extractor design: what each stage optimises, what its fixed point is, and what it costs in balanced accuracy.

Answer.

1. **Federation adds no target bias (§2).** The partition is a pure label shift (Lemma 1), so every client's data term for a class-conditional quantity points at the centralised target. The frozen extractor is, to first order, trained on the pooled objective (Corollary 2). Replacing it by a centralised class-balanced extractor changes the accuracy of a pooled-trained body with the exact readout by 0.0 / +1.1 / −1.0 points (MNIST / FashionMNIST / CIFAR-100, one seed).
2. **The generator loses rare classes through the optimiser (§3).** Coupled Adam decay on clients that lack a class shrinks its conditioning row; a count-only step ratio predicts which rows collapse (Approximation 5; 352 of 352 predicted collapses observed). Without decay, flat averaging still dilutes and mis-weights rare rows (Proposition 6); holder-weighted averaging (HWA) fixes both.
3. **The last layer is the largest loss and is fixed exactly (§4.1, §4.4).** Per-client logit corrections cannot change a single-client body (Proposition 7). A readout computed from aggregated sums equals the centralised readout (Theorem 10) and recovers 7–20 points.
4. **The body gap is closed only by synthetic data that carry their label (§4.2, §4.3).** The alternation of synthetic and real phases keeps synthetic information exactly where the client's real data are silent (Proposition 8); synthetic data without label information are an anti-discrimination penalty there (Proposition 9). An exact-statistics generator (§5) removes that regime.
5. **What remains is the long tail itself (§6.7).** On MNIST and FashionMNIST our federated pipeline loses 2.9 / 3.3 points to the long tail; the centralised pipeline loses 3.2 / 2.7.
6. **The mechanism generalises (§6.11).** Rare item embeddings in federated recommendation collapse the same way, with one difference: centralised Adam with coupled L2 already collapses the rarest items.

Contributions.

1. **A stage-by-stage account of the federated–centralised gap** for a generator-based heterogeneous FL pipeline under label skew: pure label shift (Lemma 1), first-order consistency of the extractor (Corollary 2), the reparametrisation result for per-client logit corrections (Proposition 7), the retention law of alternating phases (Proposition 8), the

logit-variance identity for uninformative synthetic data (Proposition 9) and partition invariance of sufficient-statistic readouts (Theorem 10). The individual tools are classical; their assembly into a decomposition, checked against measurements at every step, is the contribution.

2. **A diagnosis of rare-class collapse** in a FedAvg-trained conditional generator (Theorem 3, Lemma 4) with a **count-only predictor** that ranks rows before training (Approximation 5, Spearman 0.94 over 900 rows; 16 of 16 in a prospective test on SVHN), confirmed by intervention: removing the coupled decay or averaging over holders restores the rows (K15, T01), and measured round by round (K25).
3. **A decomposition of the long-tail loss** into generator, last-layer and representation parts, using eight federated readouts on every trained model, five datasets including CIFAR-100-LT (K16), and a comparison with a centralised two-stage pipeline (N7).
4. **An exact-statistics generator** (mean anchor, moment correction, herding toward the exact class embedding, an F-based switch to class rows) whose bodies beat GeFL-F's under the same readout. Its components have precedents (§7); the contribution is their use for a federated generator, explained by the signal-fraction result.
5. **Evidence with honest margins:** recalibrated baselines, the optimiser fixes, published long-tail FL losses, calibration and worst-class metrics, negative results (§6.8, Appendix B), a validation-selected re-test on new seeds against nine baselines whose pre-registered predictions partly failed (§6.6), and breadth: more clients, partial participation, other imbalance settings and random rare classes (§6.10).

2. Setting

2.1 GeFL-F

There are K clients. Client k holds data D_k and runs a head $h_{a(k)}$ of architecture $a(k)$ from a set of G architectures. GeFL-F proceeds in three stages.

1. **Feature extractor (FE) warm-up.** A small shared FE ϕ (one conv block; output $3 \times 16 \times 16$) is trained with the heads by FedAvg, then frozen.
2. **Feature generator.** A conditional VAE $G_F(h \mid y)$ (CVAE-F) is trained on $\phi(x)$ by FedAvg. It uses Adam with learning rate 10^{-3} and *coupled* weight decay $\lambda = 10^{-3}$ (paper, Table XV). The label enters through a learned linear projection whose c-th column (the *conditioning row* w_c) is the only class-specific parameter.
3. **Heads.** Each round, each client trains its head for $T_s = 1$ epoch on generated features with uniform random labels. It then trains for $T_r = 5$ epochs on its real features. Heads of the same architecture are averaged.

The paper reports `best_mean_acc`: the best test accuracy of each architecture over rounds, averaged over the architectures.

2.2 The partition is a pure label shift

Let n_{kc} be client k's count of class c, $n_c = \sum_k n_{kc}$, $N = \sum_c n_c$, $\pi_{kc} = n_{kc}/n_k$ the client prior and $\pi_c = n_c/N$ the pooled prior. The long-tailed split draws class sizes $n_c \propto \mathrm{IF}^{-c/(C-1)}$, randomly permutes each class's indices and cuts them into Dirichlet(0.5) shares (§6.1).

Lemma 1 (pure label shift). For every client k and class c with $n_{kc} > 0$, client k's class-c images are a uniformly random subset of the pooled class-c images. Hence $P_k(x \mid y) = P(x \mid y)$, and, because the extractor ϕ is shared, the same holds for features $h = \phi(x)$.

Proof. Given the counts, each contiguous chunk of a random permutation is a uniformly random subset; the top-up step that guarantees a minimum client size moves uniformly random samples and preserves exchangeability within a class. □

Consequence. For any class-conditional quantity (a generator's class row, a class mean, a readout), every client's data term points at the same target as the pooled data. Federation can then cost **weighting, variance and rate**, but not **target bias**. A bias can only come from (i) the optimiser, (ii) parameters coupled to the client's label prior (logit biases, last layers), and (iii) shared parameters trained on prior-weighted losses, which are equally prior-weighted centrally. §3 and §4 locate each.

2.3 The centralised comparator

The quantity we compare against is GeFL-F's own two-stage design trained on the pooled data: an extractor (trained federatedly, or centrally with balanced sampling), then each architecture's body trained on the **pooled** real features with the Stage-III schedule, then the same class-balanced readout. It is not end-to-end joint training; it isolates what federation costs *given* the frozen-extractor design that GeFL-F fixes. The reported metric is balanced accuracy, whose population target is the balanced risk $\mathcal{R}_{\mathrm{bal}}(\theta) = \frac{1}{C}\sum_c \mathbb{E}_{x\sim P(\cdot|c)}\ell(\theta; x, c)$.

2.4 Stage I is first-order pooled training

Stage I averages the extractor uniformly over clients, but each client runs $\tau_k \approx 5n_k/B$ local steps (five epochs; batches of size 1 are skipped). A one-round expansion of local SGD (FedNova's objective inconsistency [58]) gives the aggregate update $-\eta\bar{\tau}\nabla\tilde{\mathcal{L}} + O(\eta^2 LG\tau^2)$ with $\tilde{\mathcal{L}} = \sum_k \frac{p_k\tau_k}{\bar{\tau}}\mathcal{L}_k$.

Corollary 2 (Stage I). With $p_k = 1/K$ and $\tau_k \propto n_k$, the surrogate weights are n_k/N up to a total variation of 0.009–0.018 at $K = 10$ (0.06–0.08 at $K = 50$, 0.11–0.12 at $K = 100$; measured on every split we use). To first order the federated extractor therefore minimises the pooled loss $\sum_k \frac{n_k}{N}\mathcal{L}_k$ — but with one head per client, so its population objective is $H(Y \mid \phi(X), K) = H(Y \mid \phi(X)) - I(Y; K \mid \phi(X))$: label information that the client identity already carries is free.

Two cautions. The regime is not perturbative ($\eta\tau_k = 1$–17 at $K = 10$); for quadratic clients the exact fixed point moves toward the client-uniform average of local optima as $\eta\tau_k$ grows [59, 60]. And to second order in the class separations, the federated objective weights class pairs by $\sum_k \frac{n_k}{N}\pi_{kc}\pi_{kc'}$, which on our splits is about as long-tailed as centralised ERM's $\pi_c\pi_{c'}$ (spread of $\log_{10}$ weights 0.71–0.87 vs 0.77–0.85). So the prediction is that the federated extractor is neither better nor worse than a centralised ERM extractor, and both are below a centralised class-balanced one by however much the tiny extractor (36–390 parameters) must trade pairs off. §6.7 measures this: 0.0 (MNIST), 1.1 (FashionMNIST) and −1.0 (CIFAR-100) points.

3. Stage II: why the generator loses rare classes

3.1 The conditioning-row contraction bound

Setting. Class c's conditioning row $r_c \in \mathbb{R}^d$ ($d = 512$) is the only class-specific parameter of CVAE-F, and its data gradient is non-zero only on samples labelled c. In one round, client k starts from the global row r_c, runs S_k local Adam steps with coupled weight decay λ, and returns $r_{k,c}$. Let H_c be the m_c clients that hold class c and $p_c = m_c/K$. Write each client's change in norm as $\Delta_k = \|r_{k,c}\| - \|r_c\|$, the holders' mean gain as $\bar{G}_c = \frac{1}{m_c}\sum_{k\in H_c}\Delta_k$ and the non-holders' mean loss as $\bar{D}_c = -\frac{1}{K-m_c}\sum_{k\notin H_c}\Delta_k$.

Theorem 3 (contraction bound). Under flat FedAvg, $r_c^+ = \frac{1}{K}\sum_k r_{k,c}$,

$$\|r_c^+\| \;\le\; \|r_c\| + p_c\,\bar{G}_c - (1-p_c)\,\bar{D}_c,$$

with equality iff all $r_{k,c}$ point in the same direction. Hence the row contracts whenever

$$p_c\,\bar{G}_c < (1-p_c)\,\bar{D}_c \quad\Longleftrightarrow\quad \frac{m_c}{K} < \theta_c := \frac{\bar{D}_c}{\bar{G}_c + \bar{D}_c} \qquad (\bar{G}_c + \bar{D}_c > 0),$$

and for every $m_c < K$ when $\bar{G}_c \le 0 < \bar{D}_c$. (For $m_c = K$ the bound still allows contraction when holder updates point in different directions; T01 shows this in a few rounds for classes held by all clients.)

Proof. $\|\frac{1}{K}\sum_k r_{k,c}\| \le \frac{1}{K}\sum_k \|r_{k,c}\| = \|r_c\| + \frac{1}{K}(\sum_{k\in H_c}\Delta_k + \sum_{k\notin H_c}\Delta_k)$, and the two sums are $m_c\bar{G}_c$ and $-(K-m_c)\bar{D}_c$. □

The bound uses no property of the optimiser; it is the triangle inequality applied to the measured changes. Because $\bar{G}_c$ and $\bar{D}_c$ are measured in the same round, θ_c is an ex-post description, not a prediction: holders regrow a row that arrived shrunk, so $\bar{G}_c$ itself depends on how many non-holders there are. What the optimiser decides is the size of $\bar{D}_c$ and $\bar{G}_c$.

Lemma 4 (coupled decay under Adam). A non-holder's gradient on r_c is exactly λr_c (the same holds for a holder on steps whose batch lacks c). Adam's update is invariant to a positive rescaling of the gradient sequence up to ϵ: $\hat{m}$ and $\sqrt{\hat{v}}$ both scale with λ. So as long as $\lambda|r_{c,j}| \gg \epsilon$, a non-holder moves each coordinate of r_c toward zero by a step of order the learning rate η whatever the value of λ, and $\bar{D}_c$ does not depend on λ in that range. (Each client keeps its Adam state across rounds while its parameters are reset to the global ones, so a single step can exceed η; numerically the invariance holds for $\lambda \gtrsim 10^{-5}$ at these row norms and breaks down near $\lambda \approx 10^{-7}$.) At $\lambda = 0$ the gradient and both moments are zero and $\bar{D}_c = 0$ exactly. (Under SGD the same decay would cost only $\approx S\eta\lambda\|r_c\|$, 10^3 times less at $\eta = \lambda = 10^{-3}$.)

What this means. An earlier version of this result, "rows of classes held by fewer than half the clients shrink", was wrong: it is the special case $\bar{G}_c = \bar{D}_c$, and its per-step bound had the wrong units (a norm can move by up to $\sqrt{d}\,\eta$ per Adam step). Every holder would have to grow the row as fast as a non-holder shrinks it. Nothing forces that. A holder's data gradient fits the row; it does not push its norm up at Adam's full rate. So $\bar{G}_c \ll \bar{D}_c$, $\theta_c > 1/2$, and **tail rows contract even when most clients hold the class**. Without weight decay the non-holder term vanishes; the data step is then only diluted by m_c/K (measured in `experiments/results/T02_dilution_rates/summary.md`): rare classes learn more slowly but do not collapse.

Predictions. (i) GeFL-F's tail-to-head row-norm ratio is $\ll 1$ under a long tail, even when every class is held by more than half the clients. (ii) It is the same for every λ with $\lambda\|r\| \gg \epsilon$ and disappears at $\lambda = 0$. (iii) It worsens with K for fixed data, because more clients are non-holders of each tail class. (iv) Removing non-holders from row c's average (HWA) removes the $\bar{D}_c$ term; the row can still shrink when holder updates cancel, but no longer systematically.

3.2 Verification on the real generator (T01)

We instrumented the paper's CVAE-F and logged every client's row norms after every local round (MNIST, IF = 100, Dir 0.5, $K = 10$, 40 generator rounds, seed 0; `experiments/T01_collapse_threshold.py`, figure F01_collapse).

- *Consistency check.* Theorem 3 is satisfied in all 1,200 row-rounds of the three flat-averaging arms (mean $\|r_c^+\|$/bound = 0.983). Since the bound is computed from the same measured norms, this only confirms the instrumentation; the evidence for the mechanism is the next three points.
- *$K/2$ is the wrong threshold.* Every class is held by at least 7 of the 10 clients, so the half-the-clients rule predicts no contraction anywhere. Yet the rows of classes 7–9 shrank to 4–10% of their initial norm and those of classes 5–6 to 65–75%, while head rows (held by all 10 clients) grew by a factor 1.4. Holders gained only $\bar{G}_c/\bar{D}_c = 0.07$–$0.64$ (median per class) of what non-holders lost, so $\theta_c = 0.61$–0.93. Theorem 3 certifies 139 of the 197 contractions; the rest come from holder updates that partly cancel in direction, which the norm bound does not use.
- *Only the presence of decay matters.* $\lambda = 10^{-4}$ and $\lambda = 10^{-3}$ give the same trajectories (final tail/head ratio 0.155 vs 0.152; non-holder loss 1.49×10^{-2} per step in both). At $\lambda = 0$

the ratio stays at 0.85 and only 2 of 400 row-rounds contract.
- *HWA removes the systematic contraction.* With the same decay ($\lambda = 10^{-3}$), aggregating the rows over holders only makes the tail rows *grow* to 1.17–1.25× their initial norm (against 0.04–0.10 under flat averaging); they shrink in 1–6 of 40 rounds instead of 35–40. The final tail/head ratio is 0.72, against 0.15, because head rows grow faster still.

Other generators. The paper's DCGAN-F and DDPM-F are trained without weight decay. Then $\bar{D}_c = 0$ exactly (a non-holder's gradient on class c's parameters is zero; checked on T01 at $\lambda = 0$: largest non-holder change 0.0 in 40 rounds), and flat averaging gives $r_c^+ - r_c = p_c\times$ (mean holder update): dilution, not contraction. Rare classes still learn slowly, mainly because a holder moves the row only on steps whose batch contains the class; the predicted per-class rate $\frac{1}{K}\sum_k S_k\,[1-(1-\pi_{k,c})^B]$ matches the measured row growth on T01 (Spearman 0.85, $p = 0.002$; `experiments/T02_dilution_rates.py`). HWA speeds the rarest rows up by about 1× at $K = 10$ and 2–3× at $K = 100$. The shared parameters see the rarest class in 0.4% of the loss, the same imbalance under which centralised diffusion models and conditional GANs already lose their tail classes [CBDM, CVPR 2023; gSR, ECCV 2022], and class conditioning alone can cause mode collapse in limited-data GANs [Shahbazi et al., "Collapse by Conditioning", ICLR 2022]. The new element here is the federated optimiser–aggregation mechanism, not class-conditional collapse as such. The size of the resulting accuracy loss for DCGAN-F / DDPM-F under the long tail is not yet measured.

3.3 Measurement across seeds and datasets

The setting is IF = 100 with Dirichlet(0.5), $K = 10$, full schedule and 3 seeds (F01). *Fidelity* is the accuracy of a referee classifier, trained on real held-out features, on generated features of each class.

	MNIST row ratio	MNIST tail fidelity	FMNIST row ratio	FMNIST tail fidelity
GeFL-F	0.33 (0.09, 0.17, 0.72)	8.8	0.22 (0.55, 0.09, 0.03)	19.6
+ HWA + LA	0.80 (0.71, 0.80, 0.88)	27.2	1.01 (1.15, 1.03, 0.85)	66.2

On SVHN (Kaggle, 3 seeds) the GeFL-F ratio is 0.49, 0.32 and 0.18, against 0.99, 0.96 and 1.05 with HWA. A quick-pass ablation (E02) shows that removing weight decay, lazy decay, and HWA each restore the ratio. Prediction (iii) holds.

3.4 The intervention at full scale: fixing the optimiser vs fixing the aggregation (K15)

If coupled decay on non-holders causes the collapse, any fix that removes it should restore the rows. K15 tests the existing fixes against HWA, all with logit adjustment (LA) in the heads, on the same splits and seeds as the references (long tail, 3 seeds; tables T17 / T18):

balanced accuracy	MNIST K=10	K=50	K=100	FMNIST K=10	K=50	K=100
GeFL-F + LA (coupled decay, as in the paper)	79.2	81.4	79.5	65.2	66.5	66.6

balanced accuracy	MNIST K=10	K=50	K=100	FMNIST K=10	K=50	K=100
+ AdamW, decay 1e-3 (decoupled)	86.4	84.7	81.6	75.9	74.0	71.8
+ no decay on the class rows only	86.8	84.4	81.8	75.9	74.5	71.9
+ lazy decay (only rows in the batch)	85.1	83.9	81.7	75.8	74.2	71.6
+ HWA (coupled decay kept)	**87.9**	**86.5**	**83.1**	76.0	**75.5**	**73.3**
HWA minus best fix [95% CI]	+1.1 [−0.9, 3.1]	+1.8 [0.3, 3.2]	+1.3 [0.6, 2.0]	−0.1	+0.9 (n.s.)	+1.4 (n.s.)

- **The diagnosis holds.** Every fix restores the tail rows (tail/head norm 0.73–0.97, against 0.04–0.34 under coupled decay) and recovers 2–11 points over GeFL-F + LA (7–11 with 10 clients, less as the number of clients grows). AdamW at decay 1e-2 equals AdamW at 1e-3 (86.5 vs 86.4 on MNIST, 76.1 vs 75.9 on FashionMNIST), and the row-only fix equals the global one: the decay *term on the rows* is the cause.
- **HWA's extra gain is the dilution term.** Without collapse, a rare row still learns at a rate diluted by m_c/K (§3.2). HWA removes that too. Its advantage over the best optimiser fix is 1.3–1.8 points at K = 50 / 100 on MNIST (confidence intervals exclude zero) and points the same way on FashionMNIST; tail recall shows it more clearly (MNIST, K = 50: 74.6 vs 69.9).
- **Practical reading.** Removing coupled decay is the first fix and costs nothing (2–11 points); HWA is a refinement worth 1–2 points that grows with the number of clients. The anchored generator of §5.5 has no class rows at all, so for it the question does not arise; HWA matters for the minimal variant and for the class-row branch used on SVHN.
- **On SVHN (K18)** the same pattern holds with harder data: HWA − decay fix is +0.5 / +1.8 points at K = 50 / 100 (heads as trained; −0.1 / +0.9 with the exact readout).

3.5 A count-only predictor of collapse

Theorem 3's threshold uses the measured gains and losses of the same round. A predictor should use only what is known before training: the counts and the numbers of local steps.

Approximation 5 (count-only step ratio). Per coordinate, an Adam step has size $\eta\psi$ with $\psi = \hat{m}/\sqrt{\hat{v}} \in [-1, 1]$. A non-holder sees a constant-sign decay gradient, so $\psi = \mathrm{sign}(r_{cj})$ on each of its τ_k steps. A holder moves the row only on its $\tau_k f_{kc}$ steps whose batch contains class c (f_{kc}: hypergeometric probability), toward its target, with $|\psi| \le 1$. Under flat averaging the drift per round is

$$\Delta r_{cj} = \tfrac{\eta}{K}\Big[\psi_h \sum_{k\in H_c} \tau_k f_{kc} - \mathrm{sign}(r_{cj}) \sum_{k\notin H_c} \tau_k\Big], \qquad \kappa_c = \frac{\sum_{k\in H_c} \tau_k f_{kc}}{\sum_{k\notin H_c} \tau_k}.$$

(i) If $\kappa_c < 1$, the drift points toward zero for every coordinate and every target: the row collapses, up to an oscillation of size $O(\eta\tau/K)$. (ii) If $\kappa_c > 1$, the row settles where the holders' normalised step equals $1/\kappa_c$; for a Gaussian row model this is a shrink that falls with κ_c and with the holders' signal-to-noise ratio. Neglected: momentum carried between a holder's class-c steps and the non-stationarity of $\hat{v}$; a sparse Adam step can exceed η by up to $(1-\beta_1)/\sqrt{1-\beta_2} \approx 3.2$ [65].

Test. On T01 (MNIST, seed 0) $\kappa_c = 8.0, 7.3, 0.84, 0.31, 0.48$ for classes 5–9; their rows ended at 0.75, 0.65, 0.10, 0.04 and 0.09 of the initial norm: collapse exactly where $\kappa_c < 1$. Out of sample, over every logged GeFL-F run with coupled decay and flat averaging (MNIST and FashionMNIST at $K = 10/50/100$, CIFAR-10, CIFAR-100; 900 class rows):

κ_c	< 0.5	0.5–1	1–2	2–4	4–8	≥ 8	∞ (no non-holder)
median final / initial row norm	0.04	0.08	0.12	0.22	0.34	0.81	2.25

- Every row with $\kappa_c < 1$ ended at ≤ 0.153 of its initial norm (352 rows, no exception); Spearman(log κ_c, final norm) = 0.94.
- $\kappa_c \geq 1$ is not sufficient to survive: rows with κ_c between 1 and 8 still shrink to 0.1–0.3, as (ii) predicts when the holders' signal is weak. A calibrated equilibrium would need the per-class noise level.
- *Why centralised Adam with the same decay does not collapse here.* There $\hat{v}$ for r_c is dominated by data gradients from the batches that contain c, so decay steps are $\lambda r/\sqrt{\hat{v}} \ll \eta$. This needs the row to receive data more often than Adam's second moment forgets ($1/(1-\beta_2) = 1{,}000$ steps); every class does in GeFL-F (the rarest appears in about a quarter of the batches), but rare item embeddings in recommendation do not, and there centralised Adam with coupled L2 collapses them too (§6.11). The centralised generator itself was not run. Collapse needs three federated ingredients together: a client whose gradient on r_c is pure decay, Adam's normalisation of it, and averaging that client's row back in.
- **Prospective test (K18).** SVHN at $K = 50/100$ is a dataset the predictor had not seen; the prediction was written into this draft before the runs returned: every row with $\kappa_c < 1$ collapses (≤ 0.15 of its initial norm) under GeFL-F, and none does under HWA. Result (3 seeds, 60 rows per arm, κ_c from the per-client counts the runs log): all 16 rows with $\kappa_c < 1$ collapsed under GeFL-F (largest final norm 0.04); none did under HWA (smallest final norm 1.25); Spearman(log κ_c, final norm) = 0.99 (`docs/research/math_checks/x2_kappa_svhn_prospective.py`).
- **Round by round (K25).** Instrumenting every Stage-II round (MNIST and CIFAR-10, seed 3): for every class that some client lacks, the mean per-round shrinkage of its row on non-holders is 2–10 times the holders' growth under GeFL-F, and about 10^{-3} or less under AdamW.

3.6 Without decay: dilution and the wrong weighting

Once the coupled decay is removed, non-holders no longer move the row. What remains is a weighting and a rate problem.

Proposition 6 (row fixed points without decay; quadratic row model). Linearise the decoder at the current trunk, so that client k's class-c loss is $\frac{1}{2}\sum_{i \in D_{kc}} \|r - t_i\|_A^2$; by Lemma 1 the targets are exchangeable across clients. A local round maps $r \mapsto M_{kc} r + (I - M_{kc})\bar{t}_{kc}$ with $\bar{t}_{kc}$ the local mean. Then:

1. Flat FedAvg converges to $r^* = (\sum_{H_c}(I - M_{kc}))^{-1} \sum_{H_c}(I - M_{kc})\bar{t}_{kc}$, at a rate diluted by m_c/K. With small steps ($I - M_{kc} \propto n_{kc}$) this is the pooled estimate; with saturated local training ($M_{kc} \to 0$) it is the **unweighted** mean of the holders' local means.
2. In the saturated regime its mean squared error is $V_c = \frac{n_c}{m_c^2} \sum_{k \in H_c} 1/n_{kc} \geq 1$ times that of the pooled estimate, with equality iff all holders have equal counts.

3. HWA with weights $E(n_{kc})$ gives $V_c \approx 1$ in the saturated regime (since $E(n) \approx n$ for $n \ll 10^3$).

Proof. Non-holders contribute the identity map and drop out of the fixed-point equation; the rest is the fixed-point formula for quadratic clients and, for item 2, the variance of independent unbiased local means (Lemma 1). □

Measured. Simulated error ratios on the actual MNIST seed-0 counts match V_c to two decimals ($V_c = 2.0$–2.8 for the tail classes, 5.4–9.2 for head classes, which Dirichlet splits give holders with one or two samples). The flat per-round rate of the rarest class, relative to HWA's, explains why HWA's advantage over the decay fixes appears only with more clients: the HWA / flat rate ratio of MNIST's rarest class is 1.15 / 2.28 / 3.14 at $K = 10/50/100$, and HWA's measured gain over the best decay fix is +1.1 (n.s.) / +1.8 / +1.3 points (§3.4).

4. Stage III: heads and readout against centralised training

At $K = 10$ each architecture has one client, so head g is client g's model. A federated body differs from a body trained on the pooled data in four ways: **sample size** ($n_k \approx N/10$), **coverage** (classes the client lacks get no real gradient), **prior** (the client's label distribution), and **synthetic data** (the generator's bias and variance). This section shows which of these a correction can reach.

4.1 Per-client logit corrections are reparametrisations

Proposition 7 (exact). Let a head have a free last-layer bias b, trained by SGD without weight decay on b (as in Stage III), and let $a_k \in \mathbb{R}^C$ be any client constant: logit adjustment (LA) $a_k = \tau \log \tilde{\pi}_k$ [34], FedLC's calibration [23], or balanced softmax.

1. Training with loss $\ell(f + a_k, y)$ from (θ_0, b_0) produces exactly the trajectory of training with $\ell(f, y)$ from $(\theta_0, b_0 + a_k)$: $b_t^{\mathrm{LA}} = b_t^{\mathrm{CE}} - a_k$, and every other parameter is identical at every step.
2. FedRS [29] (logits of absent classes multiplied by α) is exactly plain cross-entropy in the variables $\tilde{w}_c = \alpha w_c$, $\tilde{b}_c = \alpha b_c$ for absent c, with the learning rate on those rows multiplied by α^2. The body follows the same trajectory.

Proof. (1) $f + a_k = Wu(h) + (b + a_k)$, and $b \mapsto b + a_k$ is a translation with identity Jacobian, so gradients agree at corresponding points; induct over steps. (2) $\alpha f_c = \tilde{w}_c^\top u + \tilde{b}_c$; the gradient in $\tilde{w}_c$ is $1/\alpha$ times that in w_c, so a step of size η on w_c is a step of $\eta\alpha^2$ on $\tilde{w}_c$, and the body sees the same logits. □

Consequences. For a single-client head trained on real data only, these four published corrections are readout corrections: they change the body only through a different initial bias or row learning rate. The identity holds to 4×10^{-16} numerically. **Measured (K17, seeds 3–5, five datasets):** heads trained on real data only with cross-entropy, LA, FedLC or FedRS have the same ER accuracy within 0.5 points on every seed and dataset (mean ER 89.96–89.97 on MNIST, 77.73–77.78 on FashionMNIST, 54.29–54.34 on SVHN). The identity breaks in two places, and both matter here: with several clients per architecture, clients with different a_k are averaged (LA then means "translate to the balanced frame before averaging"); and with a synthetic phase on unadjusted logits, the two phases see different biases (§4.2).

4.2 What the alternation keeps

Each round a client runs S_s synthetic steps, then S_r real steps.

Proposition 8 (alternation; exact for quadratic losses). Let the synthetic and real losses be quadratic with Hessians $H_s, H_r \succeq 0$ and minimisers θ_s, θ_r; let $P_s = (I - \eta H_s)^{S_s}$, $P_r = (I - \eta H_r)^{S_r}$ with $\eta < 1/\lambda_{\max}$. If $\|P_r P_s\| < 1$, the round map converges to

$$\theta^\star = \theta_r + \Omega(\theta_s - \theta_r), \qquad \Omega = (I - P_r P_s)^{-1} P_r (I - P_s).$$

When H_r and H_s commute, the synthetic weight along eigendirection i is $\omega_i = \rho_{r,i}(1 - \rho_{s,i})/(1 - \rho_{r,i}\rho_{s,i}) \le \rho_{r,i} = (1 - \eta h_{r,i})^{S_r}$.

Proof. Solve the fixed-point equation of the affine round map, as for quadratic FedAvg; the scalar form follows in a common eigenbasis. □

Reading. Along directions the client's real loss does not constrain ($h_{r,i} = 0$) the synthetic data decide ($\omega_i = 1$); along well-constrained directions the real phase forgets them; more synthetic epochs cannot raise ω_i above $\rho_{r,i}$. The directions the real data leave free are exactly those of the classes the client lacks — the ones balanced accuracy needs. For cross-entropy what matters is the real loss's curvature along the synthetic displacement, which is small once the client's real data are fit confidently. Two consequences, checked in a toy model of Stage III (3 seeds):

- **LA's second role.** Without LA, right after a synthetic phase that balanced the logits, the real phase pushes the logit of every absent class down without bound, as for logistic regression on separable data [63]. With LA the absent-class probabilities are $O(1/(n_k + C))$ and so is the push. Retention of the synthetic phase's change to absent-class rows: 0.97–0.99 with LA, 0.49–0.85 without (toy). On the real pipeline (K25, MNIST, one client every 10 rounds, counterfactual real phase from the pre-synthetic head) it is 0.96–1.0 with LA (decay fix, ours) and 0.76–0.98 without (GeFL-F): the direction holds, the size is smaller than in the toy. This is why LA matters during training even though, by Proposition 7, it is only a readout correction for real-only heads, and why its effect vanishes once ER is applied (§6.6).
- **Interleaving does not help a good generator.** Mixing synthetic and real samples in each batch minimises the joint objective instead; in the toy it gives the same ER for a faithful generator (97.2 vs 97.3) and only halves the harm of a bad one.

4.3 When synthetic data hurt: the signal fraction

Proposition 9 (label-independent synthetic data are a logit-variance penalty; exact). If the synthetic label y is uniform on $[C]$ and independent of the generated feature $\tilde{h}$, then for every logit vector f

$$\mathbb{E}_y\, \ell(f, y) = \mathrm{LSE}(f) - \bar{f} = \log C + \mathrm{KL}(u \,\|\, \mathrm{softmax} f) = \log C + \tfrac{1}{2C} \textstyle\sum_c (f_c - \bar{f})^2 + O(\|f - \bar{f}\|^3).$$

If a generated sample carries its label's information with probability s (the *signal fraction*) and is label-independent otherwise, the synthetic loss is $s\,\mathbb{E}[\ell(f(\tilde{h}), y) \mid \text{faithful}] + (1 - s)\,\mathbb{E}\,\mathrm{KL}(u \,\|\, p_f(\tilde{h}))$.

Proof. $\frac{1}{C}\sum_c(\mathrm{LSE}(f) - f_c) = \mathrm{LSE}(f) - \bar{f}$ and $\mathrm{KL}(u\|p) = -\log C + \mathrm{LSE}(f) - \bar{f}$; expand LSE at constant logits. □

Reading. At $s \approx 0$ the synthetic phase flattens the logits on the generator's support, which overlaps the real data's because the generator was fitted to real features. It is an explicit anti-discrimination regulariser, and by Proposition 8 it survives exactly in the absent-class directions. The prediction is a sign change: the synthetic phase hurts the body (relative to real-only heads) when s is near zero and helps once the faithful term dominates.

Test (K17, seeds 3–5). We estimate $s = (a - 1/C)/(1 - 1/C)$ from the held-out referee's accuracy a on generated features, and measure the synthetic gain as ER(arm) − ER(heads trained on real data only), same seeds and splits:

s / synthetic gain (ER points)	GeFL-F	GeFL-F + AdamW + LA	Ours ($T_s = 10$)
MNIST	0.31 / +1.0	0.53 / +2.6	0.97 / +4.7
FashionMNIST	0.50 / +2.1	0.72 / +2.8	0.87 / +3.4
SVHN	0.36 / +9.2	0.46 / +9.9	0.46 / +11.1
CIFAR-10	0.034 / **−0.8**	0.074 / +0.1	0.23 / +1.5
CIFAR-100	0.002 / **−2.2**	0.010 / **−1.2**	0.088 / +1.8

- Within every dataset the gain increases with s, and it changes sign between $s \approx 0.03$ and 0.07 on the CIFARs. On CIFAR-100, GeFL-F's generator is worse than no generator at all; ours turns the synthetic phase into a gain.
- The scale differs across datasets: SVHN gains most at moderate s because its real-only bodies are the weakest (ER 54.3).
- In the toy model, class-agnostic synthetic data lower ER from 85.7 (real only) to 77.9 at $T_s = 1$ and 68.4 at $T_s = 10$ — more epochs, more harm — while a faithful generator raises it to 96.0 / 97.3.
- **CSL** (§5.3) changes the target to $(1 - \beta)e_y + \beta\bar{p}(\tilde{h})$. At $s = 0$ its loss is $(1 - \beta)$ times the penalty plus distillation of the cross-architecture ensemble on the generator's support, a feature-dependent target. In the toy it turns the harm into a gain (68.4 → 90.2 at $T_s = 10$).

The signal fraction is therefore the quantity a federated generator must raise. Rows that collapse (§3) carry no class information ($s \to 0$); the exact-statistics generator of §5 conditions on the exact class mean and corrects the class moments, which is why its s is 7 and 44 times GeFL-F's on CIFAR-10 and CIFAR-100.

On the real pipeline (K25, CIFAR-10, seed 3). The logit spread on generated features after the synthetic phase, relative to before it, settles at about 0.5–0.6 for GeFL-F and 0.7–0.8 for the decay fix and for ours (rounds ≥ 50); in absolute terms GeFL-F's logits on its own synthetic features end with a spread of about 1, ours about 7. The phase flattens the logits most where the generator carries the least label information, as Proposition 9 says.

A gate that follows from Proposition 9 (K21). If the server can estimate s_c per class without held-out data, it can withhold or relabel the synthetic data of classes with $s_c \approx 0$. We estimate $\hat{s}_c$ from the agreement of the class-balanced LDA on the frozen features (exact sums, Theorem 10) with the conditioning label of generated samples, normalised by the same LDA's recall on real data, and either keep only classes with $\hat{s}_c \geq 0.1$ (SGATE) or use the consensus label with weight $1 - \hat{s}_c$ (SCSL). Seeds 3–5, paired with K17, ER:

- For the decay-fixed GeFL-F, whose synthetic phase hurts on the CIFARs, the gate helps as predicted: SCSL +0.77 on CIFAR-10 and SGATE +1.31 on CIFAR-100 (3 of 3 seeds each), which moves its synthetic gain over real-only heads to ≥ 0 on both (pre-registered Q1: holds). On MNIST / FashionMNIST, where s is high, SGATE changes ER by −0.4 / −0.1.
- For ours it changes nothing useful: $\hat{s}_c \approx 0.93$–1.0 for every class, so SGATE keeps all classes (its +0.2 / +0.3 on CIFAR-100 / CIFAR-10 is a replicate of ours under new code), and SCSL, which then sets the consensus weight near 0, removes CSL and costs 0.2–1.9 points (Q2: fails). The estimator is the problem: moment correction puts the generated class means exactly on the true ones, which is all an LDA sees, whereas the referee (a trained network) finds $s = 0.09$ on CIFAR-100. Across all K21 runs Spearman($\hat{s}_c$, referee s_c) = 0.48 (Q3, ≥ 0.6: fails; 0.87 on MNIST, 0.33 on CIFAR-10).

4.4 The readout is partition-invariant

Theorem 10 (partition invariance of sufficient statistics). Let an estimator be a function $\Psi(S)$ of $S = \sum_k \sum_{i \in D_k} T(x_i, y_i)$ with T fixed in advance. Its value is the same for every partition of the pooled data into clients and equals the centralised estimator. In particular every exponential-family maximum-likelihood estimate is computed exactly by one round of secure aggregation.

Proof. Addition is associative and commutative. □

The theorem is trivial; its use is to sort the pipeline. ER (class-balanced LDA from per-class n_c, $\sum e$ and $\sum ee^\top$ of each architecture's embeddings), Fed3R, FedNCM, all-client CCVR, the gate statistic F, and the targets of the mean anchor, moment correction and herding are partition-invariant: they equal their centralised values for the same frozen networks. The extractor, the generator trunk and the bodies, trained by FedAvg or local SGD, are not; neither are LA / FedLC / FedRS (client-specific by design) or the CSL and BBC targets (they depend on trained models). Consequences:

- **ER equals centralised LDA on the same bodies.** The federated last layer is no longer a source of loss relative to centralised training of those bodies; what remains is LDA's model error. Measured (K16): ER, all-client CCVR and a pooled-real refit of the last layer agree within 1–2 points on all five datasets.
- **ER accuracy measures the body.** It never uses the trained last layer, only the class means and the within-class covariance of the embeddings over all clients, so it is a direct measure of the body's class geometry. A head trained on one client is the extreme of minority collapse [62] — absent classes have imbalance ratio ∞ and their last-layer rows carry only what the synthetic phase put there — and ER sidesteps it.
- **The statistical floor.** The exact class mean has error V_c/n_c centrally as well; relative to the typical squared distance between class means ($2F\bar{V}$) this is $1/(2Fn_c)$: 0.07 (MNIST), 0.04 (FashionMNIST), 0.07 (CIFAR-10), 0.48 (CIFAR-100) and about 1.7 (SVHN) for the rarest class. On SVHN a tail class's exact mean is uninformative even centrally — a second, independent reason, besides Proposition 12, why the mean anchor fails there. No aggregation rule can beat this floor.

5. Method

5.1 HWA: holder-weighted aggregation of conditioning rows

Let $n_{k,c}$ be client k's count of class c, and $E(n) = (1 - \beta^n)/(1 - \beta)$ the effective number of samples (Cui et al., 2019), with $\beta = 0.999$. The server aggregates

$$w_c \leftarrow \frac{\sum_k E(n_{k,c})\, w_c^{(k)}}{\sum_k E(n_{k,c})}, \qquad \text{all other parameters by FedAvg.}$$

- **Effect on §3.1.** Non-holders have $E(0) = 0$, so the decay-only copies are excluded. By the same triangle inequality, $\|r_c^+\| \le \|r_c\| + \sum_{k \in H_c} \omega_k \Delta_k$ with $\omega_k \propto E(n_{k,c})$: the $-(1 - p_c)\bar{D}_c$ term of Theorem 3 disappears for every class with at least one holder. HWA does not force the row to grow; it leaves it to the holders, weighted toward those with the most samples.
- **Why effective number (a heuristic).** A holder's row estimate has variance that falls with its sample count but plausibly saturates (correlated samples, a finite number of local steps). $E(n)$-weighting is a convenient interpolation between the two limits, raw counts n ($\beta \to 1$) and uniform-over-holders ($\beta \to 0$); we do not derive it. K02 ablates both limits: the three rules are within one point of each other (counts 89.97 / 76.45, uniform 89.02 / 76.33, $E(n)$ 90.15 / 76.51 on MNIST / FMNIST; 2 seeds), so what matters is excluding non-holders, not the exact weights.

- **Do-no-harm.** When every client has the same class counts, HWA *equals* FedAvg. Random IID splits give nearly equal counts. On SVHN IID, HWA + LA scores 0.18 points below GeFL-F (75.64 vs 75.82), a small but consistent cost.
- **Privacy.** The server needs $\sum_k E(n_{k,c}) w_c^{(k)}$ and $\sum_k E(n_{k,c})$, which are sums. Under secure aggregation it learns neither any client's row nor any client's histogram. We also test Laplace-noised counts ($\varepsilon \in \{10, 1, 0.1\}$; K02).

5.2 LA: heads trained with the client's own prior

Each head trains on real features with logits $f(h) + \tau \log \pi_k$, where π_k is client k's smoothed label distribution and $\tau = 1$. Testing uses plain $f(h)$. Cross-entropy on prior-adjusted logits is Fisher-consistent for the balanced error (Menon et al., 2021). Using the *client's own* prior means each local update estimates the balanced risk for that client's label shift, so averaged heads are balanced as well. The prior never leaves the client. In IID data π_k is uniform and LA is a no-op.

What LA can and cannot do follows from §4.1–4.2. For a single-client head trained on real data alone it is an exact reparametrisation of the last-layer bias (Proposition 7), so it changes nothing that a readout does not also fix; with ER its effect vanishes (§6.6). Its role during training is to keep the real phase from erasing what the synthetic phase taught about the classes the client lacks: without it the real phase drives those logits down without bound, with it they receive almost no gradient (Proposition 8 and its reading). With several clients per architecture it also puts every client's head in the balanced frame before averaging.

5.3 CSL: consensus soft labels for synthetic features

A generated feature $\tilde{h} \sim G_F(\cdot \mid y)$ is not always a clean example of y. Its true posterior $p(\cdot \mid \tilde{h})$ differs from onehot(y) by the generator's error. The minimiser of expected cross-entropy is that posterior, so one-hot training on generated features injects label noise. The server already holds every architecture's aggregated head. For a shared synthetic pool drawn from a public seed it computes

$$t(\tilde{h}) = (1 - \beta)\,\mathrm{onehot}(y) + \beta\,\frac{1}{G}\sum_{g=1}^{G} \mathrm{softmax}\, h_g(\tilde{h}), \qquad \beta = 0.5,$$

and clients use t as the synthetic-phase target. The G heads have different architectures and are trained on different clients' data. Averaging their errors lowers variance by up to $1/G$ for uncorrelated errors, and the one-hot term bounds the bias that a shared ensemble error would introduce. This is the bias–variance reason why $\beta = 0.5$ beats $\beta = 1$ (§B.4). The bias–variance view of distillation from teachers that approximate the Bayes posterior is due to Menon et al. (ICML 2021); CSL applies it to generated features labelled by a cross-architecture ensemble.

Protocol and cost. GeFL-F aggregates heads only within an architecture, so the server must evaluate one head per architecture on the pool. In our implementation it holds the per-architecture aggregated heads (as the server of a model-heterogeneous FedAvg does when it aggregates them); a deployment that does not keep heads at the server needs each architecture group to upload its predictions on the pool (a $P \times C$ sum per group per round, computable under secure aggregation). CSL therefore needs a message GeFL-F does not: head parameters or pool predictions per round. The server broadcasts the pool seed and the $P \times C$ soft labels (about 24 KB per round here). CSL uses no real data.

5.4 Why consensus labels make more synthetic data useful (Proposition 11)

Stylised model. A head estimates a parameter $\theta^\star$ from n_r real samples and a random fraction $\lambda \in [0, 1]$ of n_s synthetic samples, each at unit weight (a subsampling model; more synthetic epochs raise the number of synthetic samples used). This is a toy model of the head's training, not

of SGD on the actual network. Real samples are unbiased with per-sample variance σ^2. Synthetic samples carry the same variance, but their labels are biased by $b(t)$, the distance between the label target t and the true posterior of the generated feature. The weighted estimator's mean-squared error is

$$\mathcal{E}(\lambda) \;=\; \frac{\sigma^2}{n_r + \lambda n_s} \;+\; \Big(\frac{\lambda n_s}{n_r + \lambda n_s}\Big)^2 b^2.$$

Proposition 11. For $b > 0$, $\mathcal{E}$ has a unique minimiser $\lambda^\star(b)$ in $[0, 1]$ (or the boundary $\lambda^\star = 1$). It is non-increasing in b, and $\lambda^\star \to 1$ as $b \to 0$.

Proof. Write $w = \lambda n_s/(n_r + \lambda n_s) \in [0, w_{\max}]$, with $w_{\max} = n_s/(n_r + n_s)$. This is an increasing bijection of λ. Since $n_r + \lambda n_s = n_r/(1 - w)$, the error becomes

$$\mathcal{E}(w) = \frac{\sigma^2(1 - w)}{n_r} + w^2 b^2.$$

This is a convex parabola in w, minimised at $w^\star = \sigma^2/(2n_r b^2)$ and clipped to $[0, w_{\max}]$. $w^\star$ is decreasing in b, and λ is increasing in w. So $\lambda^\star$ is non-increasing in b. As $b \to 0$, $w^\star \to \infty$, so the clip gives $w^\star = w_{\max}$, that is $\lambda^\star = 1$. □

Reading. With hard labels, b is the generator's label error, which no amount of synthetic data averages away. The optimal synthetic share is then small, and adding synthetic epochs ($T_s\uparrow$, which acts like raising λn_s) past it *hurts*, or at best does nothing. This is the paper's Fig. 10. CSL's target $t = (1 - \beta)e_y + \beta\,\bar{p}$ moves t toward the posterior, which lowers b and pushes $\lambda^\star$ up, so the same extra epochs now help. The proposition predicts an *interaction*, not a main effect: the gain from more synthetic epochs should be larger with CSL than without it. That is exactly what E16 measures (FMNIST IID: +0.22 from $T_s = 1$ to 5 without CSL, +0.83 with it). It also explains why the budget helps more under the long tail once HWA has made tail features faithful, since a lower generator error is a lower b as well.

5.5 Anchoring every stage to exact statistics

HWA repairs a FedAvg-trained generator. The diagnosis of §3, together with the measurements of §6, points to a cleaner design. Wherever the data decides the result, use a statistic that secure aggregation computes **exactly** — sums over clients of per-class quantities — instead of a parameter that FedAvg *estimates*. Every statistic below costs one upload per client and reveals only federation-level sums.

PC-VAE: class identity from the exact class mean.

- *Prototypes.* The server computes $m_c = \sum_k \sum_{i \in k, y_i = c} h_i \,/\, \sum_k n_{kc}$.
- *Decoder.* It models only the within-class residual, $\tilde{h} = \mathrm{ReLU}(m_y + \mathrm{dec}(z, m_y))$, and receives m_y through a class-*shared* projection.
- *Encoder.* It sees m_y as an extra channel.
- *Consequence.* There is no class-specific parameter, so the collapse recursion of §3 has nothing to act on. Tail classes borrow within-class variation from every class through the shared decoder. A linear decoder recovers a Gaussian class-conditional model. The anchor does **not** guarantee that samples carry their class: the decoder can ignore m_y, and when class means are close it does (next paragraph, and SVHN in §B.11).

What a mean-anchored generator can express. The anchor removes the collapse, but it also limits what the generator can say about a class. Write the class-c sampler as $X_c = \mathrm{ReLU}(m_c + g(Z, Am_c))$, with $Z \sim \mathcal{N}(0, I)$ the same for every class, A the class-shared projection, and g L-Lipschitz in its second argument.

Proposition 12. For any two classes, $W_1(G_c, G_{c'}) \le (1 + L\|A\|_2)\,\|m_c - m_{c'}\|$.

Proof. Couple both classes through the same Z. Since ReLU is 1-Lipschitz, $\|X_c - X_{c'}\| \le \|m_c - m_{c'}\| + \|g(Z, Am_c) - g(Z, Am_{c'})\| \le (1 + L\|A\|_2)\|m_c - m_{c'}\|$. Taking expectations bounds W_1. □

- *What it means for a head.* If a head's logit differences are K-Lipschitz, the synthetic data can separate c from c' by an expected margin of at most $K(1+L\|A\|_2)\|m_c - m_{c'}\|$, however different the real classes are. MC adds at most the difference of the exact spreads, and KH only re-weights candidates drawn from G_c.
- *The quantity that decides.* In units of the within-class spread the anchors are, on average, $\sqrt{2F}$ apart, where $F = \overline{\|m_c - \bar{m}\|^2} \,/\, \overline{V_c}$. The server computes F from the sums it already receives. In the frozen feature space where the gate computes it (logged as `class_sep_F` in every run), F is 0.40–0.71 (FashionMNIST), 0.25–0.36 (MNIST), 0.055–0.078 (CIFAR-10) and **0.005–0.013 (SVHN)**, the range being over splits and settings.
- *Why training does not escape it.* On SVHN the anchors differ by about 10–15% of the within-class spread ($\sqrt{2F}$ with $F \approx 0.005$–0.013). Reproducing class differences of the order of the spread needs a decoder roughly 6–10 times more sensitive to the anchor than to its noise. The ELBO offers a cheaper route: the encoder sees x, so it can carry the label in z for a KL cost of order $\log C$ nats per sample, small against the reconstruction term. Samples drawn from the prior are then class-agnostic. This is what we measure (§B.11): after KH, tail fidelity is 24% on SVHN, against 36% for the CVAE + HWA and 0.9 on MNIST.
- *The fix.* Free class rows r_y inside the decoder, $g(Z, Am_y + r_y)$, remove the bound: the separation $\|r_c - r_{c'}\|$ is then a learned parameter, and HWA keeps it from collapsing. This is the hybrid generator PCR.

MC: the minimum-displacement moment correction. The server also knows each class's exact spread $V_c = \mathbb{E}\|h - m_c\|^2$, from the sums of $\|h\|^2$. Let $\tilde{m}_c$ and $\tilde{V}_c$ be the mean and spread of the generator's class-c samples. MC maps $x \mapsto \mathrm{ReLU}(m_c + \sqrt{V_c/\tilde{V}_c}\,(x - \tilde{m}_c))$.

Proposition 13. Among **all** maps T with $\mathbb{E}\,T(X) = m_c$ and $\mathbb{E}\|T(X) - m_c\|^2 = V_c$ (with X the generator's class-c sample), the map $T^\star(x) = m_c + \sqrt{V_c/\tilde{V}_c}\,(x - \tilde{m}_c)$ minimises the expected displacement $\mathbb{E}\|T(X) - X\|^2$.

Proof. Let $Y = T(X)$. Since $\mathbb{E}[Y - m_c] = \mathbb{E}[X - \tilde{m}_c] = 0$, $\mathbb{E}\|Y - X\|^2 = V_c + \tilde{V}_c + \|m_c - \tilde{m}_c\|^2 - 2\,\mathbb{E}\langle Y - m_c,\, X - \tilde{m}_c\rangle$. By Cauchy–Schwarz, the last expectation is at most $\sqrt{V_c\tilde{V}_c}$, with equality iff $Y - m_c = \sqrt{V_c/\tilde{V}_c}\,(X - \tilde{m}_c)$. □

In the idealised setting of Proposition 13, MC matches the two moments the generator is measurably wrong about while changing the samples as little as possible. In practice two things break exactness: the final ReLU (which projects onto the non-negative orthant where FE features live) changes both moments again, and $\tilde{m}_c, \tilde{V}_c$ are plug-in estimates from a finite batch of generated samples. MC is therefore an approximate moment match.

MC-S: the same correction with the exact class covariance. Second moments are also sums.

- *Release.* The pooled within-class covariance (the release FSG already uses) defines a top-k principal basis B, which the server broadcasts. Clients then upload class sums of $z = B^\top h$ and of $zz^\top$ (Ck^2 numbers; $k = 256$).
- *Target.* For each class this gives the exact class covariance Σ_c in that subspace. For tail classes it is shrunk toward the pooled covariance with prior strength k (an inverse-Wishart prior), and rescaled to keep the class's exact in-subspace variance. Only the *shape* is borrowed.
- *Map.* Let $\tilde{\Sigma}_c$ be the generator's class-c covariance in the subspace. MC-S maps $x \mapsto \mathrm{ReLU}(m_c + BA_cB^\top(x - \tilde{m}_c) + s_c(I - BB^\top)(x - \tilde{m}_c))$, with $A_c = \tilde{\Sigma}_c^{-1/2}\big(\tilde{\Sigma}_c^{1/2}\Sigma_c\tilde{\Sigma}_c^{1/2}\big)^{1/2}\tilde{\Sigma}_c^{-1/2}$.

The scalar s_c restores the exact spread outside the subspace, as in MC.

- *Practical details.* $\tilde{\Sigma}_c$ is estimated from generated samples and regularised (shrinkage toward its diagonal and eigenvalue clipping) before the matrix square roots; the map is applied before the ReLU, so the same caveats as for MC apply. The release adds a $D \times D$ pooled covariance (5.0–28.8 MB depending on the feature dimension), which matters for the cost comparison in §B.12.

Proposition 14. Among all maps T with $\mathbb{E}\,T(X) = m_c$ and $\operatorname{Cov} T(X) = \Sigma_c$ (with X the generator's class-c sample, $\tilde{\Sigma}_c \succ 0$), the map $T^\star(x) = m_c + A_c(x - \tilde{m}_c)$ minimises $\mathbb{E}\|T(X) - X\|^2$. The minimum is the squared Gelbrich distance $\|m_c - \tilde{m}_c\|^2 + \operatorname{tr}\Sigma_c + \operatorname{tr}\tilde{\Sigma}_c - 2\operatorname{tr}\big(\tilde{\Sigma}_c^{1/2}\Sigma_c\tilde{\Sigma}_c^{1/2}\big)^{1/2}$.

Proof. Let $Y = T(X)$ and $K = \mathbb{E}[(Y - m_c)(X - \tilde{m}_c)^\top]$. Then $\mathbb{E}\|Y - X\|^2 = \|m_c - \tilde{m}_c\|^2 + \operatorname{tr}\Sigma_c + \operatorname{tr}\tilde{\Sigma}_c - 2\operatorname{tr} K$. The joint covariance $\left[\begin{smallmatrix} \Sigma_c & K \\ K^\top & \tilde{\Sigma}_c \end{smallmatrix}\right]$ is positive semi-definite, so $K = \Sigma_c^{1/2} R\, \tilde{\Sigma}_c^{1/2}$ with $\|R\|_2 \le 1$. Hence $\operatorname{tr} K \le \|\tilde{\Sigma}_c^{1/2}\Sigma_c^{1/2}\|_* = \operatorname{tr}\big(\tilde{\Sigma}_c^{1/2}\Sigma_c\tilde{\Sigma}_c^{1/2}\big)^{1/2}$. $T^\star$ attains the bound. It gives $K = A_c\tilde{\Sigma}_c$, whose trace is exactly this value, and $\operatorname{Cov} T^\star(X) = A_c\tilde{\Sigma}_c A_c = \Sigma_c$. □

In the subspace-plus-complement form, the displacement splits orthogonally. The in-subspace part is bounded by Proposition 14 and the complement part by Proposition 13, and the MC-S map attains both bounds at once. MC-S is therefore the minimum-displacement map that achieves the exact class mean, the exact class covariance in the principal subspace and the exact spread outside it. The bound itself is classical (Gelbrich, 1990; Olkin and Pukelsheim, 1982). What is new is using exact federated second moments to correct a federated generator.

Why it matters. By Proposition 12, a mean-anchored generator cannot tell classes apart when their means coincide. MC-S gives each class its exact covariance, so it can, wherever the covariances differ.

- *Information in exact statistics* (pixel space, balanced accuracy, classifiers built only from the statistics):

	nearest class mean	LDA (FSG's model)	QDA (class covariances)
MNIST	80.4	**87.2**	79.8
FashionMNIST	67.8	**81.3**	74.7
CIFAR-10	28.2	39.3	**51.4**
SVHN	10.3	20.6	**54.4**

- *Proxy test.* A class-agnostic generator (the SVHN failure mode, simulated by drawing from the pooled data) is corrected per class. A head is trained only on the corrected samples and tested on real data:

balanced / long tail	MC (mean + spread)	**MC-S (mean + covariance)**	real data, same counts
MNIST	65.2 / 63.4	**89.7 / 78.1**	95.1 / 87.1
CIFAR-10	22.6 / 23.0	**35.2 / 30.5**	43.7 / 33.7
SVHN	10.6 / 12.2	**43.9 / 28.6**	69.6 / 51.9

MC transfers nothing on SVHN. MC-S transfers 63% of real-data accuracy in the balanced case and 55% under the long tail. The full-pipeline test is K11 (§B.11).

KH: kernel herding toward the exact class embedding.

- *Embedding.* With bounded random Fourier features ϕ, the server receives the exact class mean embeddings $\mu_c = \sum_k \sum_{i \in k, y_i = c} \phi(h_i)/n_c$ in one upload.

- *Selection.* For each class, from $4n$ calibrated candidates, KH greedily selects n without replacement so that $\|\frac{1}{n}\sum_t \phi(x_t) - \mu_c\|$ stays small (a herding-style greedy selection; it is not the Frank–Wolfe kernel-herding iteration, which samples from the whole space and may repeat points).
- *Motivation, not a guarantee.* For a head whose loss lies in the RKHS with norm at most B, $|\mathbb{E}_{G_c}\ell - \mathbb{E}_{P_c}\ell| \le B\,\mathrm{MMD}(G_c, P_c)$. The premise does not hold for the ReLU heads we train, so this only motivates the target. Empirically KH lowers the MMD but does not improve train-on-synthetic, test-on-real accuracy, and its effect on the final accuracy is small and not significant (§B.11).

BBC: balanced bias calibration on the anchored generator.

- *Fit.* After training, the server fits C logit offsets b_g per head g so that the head predicts every class equally often on class-balanced synthetic data. This is the prior-correction fixed point $b \leftarrow b - \log(C\,\bar{q}(b))$ of Saerens et al. (2002) in logit form.
- *Gate.* It is switched on only when the exact global class counts are imbalanced (max/min > 2). In our settings the ratio is near 1 (IID) or near the imbalance factor (long tail), so other thresholds in a wide range would make the same decisions, but no other threshold was run. It uses no test or held-out data.
- *Role.* It corrects the residual bias that LA's prior correction leaves when tail classes are also intrinsically hard (FashionMNIST's shirt and coat), without tuning LA's temperature.

VCT: vicinal consensus transfer (heads). The federation already knows more than any of its heads. On CIFAR-10 the ten heterogeneous heads *together* reach 69.8% while the average head reaches 60%. Consensus labels on synthetic features (CSL) transfer little of that gap (+0.75), because synthetic features sit off the real data. Image-space augmentation helps for a different reason, regularising each model on its own data (GeFL + MixUp: 62.67).

- *Rule.* In its real-data epochs, client k mixes every real feature with a partner from the round's synthetic pool: $\tilde{x} = \lambda x + (1-\lambda)x_s$, with $\lambda = \max(u, 1-u)$ and $u \sim \mathrm{Beta}(1,1)$.
- *Target.* $\lambda\, e_y + (1-\lambda)\, q(x_s)$, where $q(x_s)$ is the partner's consensus soft label from all architectures.
- *Long tail.* LA uses the mixed sample's own label distribution, $\lambda\,\pi_k + (1-\lambda)\,u_C$.
- *Disclosure.* Nothing new is disclosed. The pool and its consensus labels are the CSL release, and the real features never leave the client.

Proposition 15 (MixUp is determined by class moments to second order). Let the cross-entropy MixUp objective use partners $(x', y') \sim Q$. Expand it to second order in $\delta = (1-\lambda)(x' - x)$. It then depends on Q only through the class prior π_c, the class means μ_c and the class covariances Σ_c.

Proof. Cross-entropy is linear in the target, so the objective is $\mathbb{E}[\lambda\,\ell(f(\tilde{x}), y) + (1-\lambda)\sum_c y'_c\,\ell(f(\tilde{x}), e_c)]$. Expand $\ell(f(x+\delta), t) = \ell(f(x), t) + \nabla_t^\top \delta + \frac{1}{2}\delta^\top H_t \delta + O(\|\delta\|^3)$. Given λ, the expectation over Q needs $\mathbb{E}[y'_c] = \pi_c$, $\mathbb{E}[y'_c\,\delta] = (1-\lambda)\,\pi_c(\mu_c - x)$ and $\mathbb{E}[y'_c\,\delta\delta^\top] = (1-\lambda)^2\,\pi_c\,[\Sigma_c + (\mu_c - x)(\mu_c - x)^\top]$. The first term's moments are the π-mixtures of these. □

- *Numerical check.* With skewed, non-Gaussian classes, the MixUp loss with real partners and with Gaussian partners that match only (π_c, μ_c, Σ_c) agree to within 0.0003–0.0010 (losses near 2.2). Partners that match only the means are off by up to 0.069.
- *Consequence.* Partners from the MC-S-corrected generator, which match the exact class means and covariances, give each client the MixUp regulariser of the *whole federation's* data. No client sees another's data. Under the long tail this matters most: a client cannot mix toward classes it does not hold, but the synthetic partners cover every class. The consensus part of the target is not moment-determined. It is the distillation of the ensemble along the path from the client's real data into the federation's knowledge.

KME-Gen: the extreme case. Train the generator *at the server only*, by MMD against the exact class embeddings μ_c, with no federated generator training at all. This is the DP-MERF construction (Harder et al., 2021) applied to federated feature sums; we include it as a reference point, not as a contribution.

- *Partition invariance.* μ_c equals the centralised embedding for every partition, so collapse, dilution and client drift cannot occur.
- *Formal privacy.* Since $\|\phi\| \le \sqrt{2}$, Gaussian-mechanism DP on the single release covers the whole synthetic channel by post-processing.
- *Cost.* One upload of about 0.3 MB replaces 100 rounds of generator FedAvg.
- *Result.* It beats GeFL-F by 13–16 points, but a learned decoder anchored by PC + MC + KH is better (§B.11).

5.6 Exact readout (ER): the last layer from exact sums

After training, the server replaces each architecture's last layer by the class-balanced linear discriminant computed from exact statistics of that architecture's embeddings $e = \text{body}_g(h)$:

- every client passes its real features through the body of architecture g and uploads per-class counts n_c, sums $\sum e$ and the sum $\sum ee^\top$ (secure-aggregatable);
- the server forms class means μ_c and the pooled within-class covariance S, shrinks it, $S_\gamma = (1-\gamma)S + \gamma\,\text{tr}(S)/d\,I$ with $\gamma = 0.1$, and sets $w_c = S_\gamma^{-1}\mu_c$, $b_c = -\frac{1}{2}\mu_c^\top S_\gamma^{-1}\mu_c$ (uniform class prior: balanced error).

Scope and cost. With "all clients", every architecture's body (small: up to about 10^5 parameters) is broadcast once to every client, and each client uploads up to about 5×10^5 numbers per architecture ($d \le 1024$, symmetric $\sum ee^\top$): one extra round. With "own clients" only the architecture's clients contribute, as in published per-model protocols (CCVR, Fed3R, FedNCM); at K = 10 that is one client, and those readouts are much weaker for every method (§6.3). ER is the LDA member of a family of closed-form readouts (CCVR, Fed3R, FedNCM); §6.3 reports all of them.

Once, not during training. Recomputing the readout every round and training only the bodies under it (ERL) was tested and does not help (§6.8).

5.7 The method

Ours = anchored generator + LA/CSL heads + exact readout.

- *Generator.* $F = \overline{\|m_c - \bar{m}\|^2}/\overline{V_c}$ is computed from the released sums. $F \ge 0.05$: the PC-VAE, conditioned on the exact class mean (no class rows). $F < 0.05$: the paper's CVAE-F with class rows protected by HWA (or, equivalently for the collapse, by decoupled decay; §3.4). Implemented as the `GATE` component, read at run time. The threshold was set after the SVHN failure; measured F is 0.005–0.013 on SVHN, 0.054–0.080 on CIFAR-10, 0.18–0.25 on CIFAR-100 and at least 0.25 on MNIST / FashionMNIST, so CIFAR-10 lies close to it (its row branch is tested in K17).
- *Sampling.* MC (exact mean and spread), then KH (herding toward the exact class embedding).
- *Heads.* LA by the client's own prior, CSL on synthetic features; $T_s = 10$ on MNIST / FashionMNIST and 1 elsewhere (chosen on test results so far; K17 chooses it on validation data).
- *Readout.* ER after training (§5.6); BBC is an alternative that needs no extra round but is weaker.

The minimal variant for an existing GeFL-F deployment is HWA (or decoupled decay) + LA + CSL, with the paper's generator, plus ER.

6. Experiments

6.1 Protocol

We follow the paper's appendix exactly.

- **Models.** FE per Table XIV (CIFAR-10: 10 channels). Heads CNN-1…10 (Tables XXI–XXIII). CVAE-F per Table XX.
- **Optimisation.** SGD with learning rate 0.1 and momentum 0. Rounds $T_{FE}/T_{KA}/T_{TN} = 20/100/50$ (MNIST, FMNIST); SVHN and CIFAR-10 per the paper.
- **Data.** Fraction 0.1 (0.5 for CIFAR-10). Client-to-architecture assignment is contiguous. With $K = 10$ clients and 10 head architectures, **each architecture has exactly one client**, so "aggregating heads within an architecture" is a no-op at $K = 10$ and only matters at $K = 50, 100$.
- **Long tail.** $n_c \propto \mathrm{IF}^{c/(C-1)}$ with the total data budget fixed, and Dirichlet(0.5) client shares. Test sets are balanced. The rare classes are always classes 6–9 (the order is not permuted across seeds). With $K = 10$ the rarest classes are still held by 6–8 of the 10 clients. On CIFAR-10 the subsampled training set (18,485 images) has a realised imbalance of 48:1, not 100:1, because the rarest classes are capped by the fraction.
- **Metrics.** Final-round balanced accuracy is the primary metric. We also report tail recall (classes 6–9), worst-class recall, and the paper's `best_mean_acc`, which takes the best round of the test-set mean accuracy and is therefore optimistic; wherever we report it we give the final-round value beside it.
- **CIFAR-100-LT.** CIFAR-10's extractor and schedule, heads with 100 outputs, fraction 0.22: at IF = 100 the federation holds 10,926 images, 480 in the largest class and 5 in the smallest (the standard CIFAR-100-LT shape); head / medium / tail are the first 30 / next 30 / last 40% of classes.
- **Readouts.** Every trained model is scored as trained and under BBC, ER, Fed3R, FedNCM and CCVR (each with all clients or the architecture's own clients) and the pooled-real oracle (K16, K17).
- **Further metrics.** Macro-F1, G-mean of class recalls (each floored at 0.01), worst architecture, ECE, NLL, Brier score, and accuracy re-weighted to each head's own clients' label distribution.
- **Selection.** Until K17 there was no validation split: T_s, β, τ and the gate threshold were chosen on the test sets we report (§8). From K17 on, the 2,000 held-out training images that no client holds serve as a validation set, and settings are chosen on them only.
- **Statistics.** 3 seeds unless marked; each seed fixes the data split and the initialisation for every method, but GPU training is not bit-deterministic (cuDNN autotuning was on), so a repeated seed does not reproduce exactly. p-values come from two-sided paired t-tests over seeds, without multiplicity correction; with 2–3 seeds they are indicative only.

6.2 Is our GeFL-F the paper's GeFL-F?

Setting	Paper	Authors' code	Ours
MNIST IID, K=10	95.47	96.07	96.02 ± 0.04
MNIST IID, K=50	95.04	–	95.16
MNIST IID, K=100	94.63	–	94.61
FMNIST IID, K=10	83.14	–	82.67 ± 0.31
MNIST LT, seed 0 (final balanced)	–	75.84	73.92

Under the long tail our baseline is 1.9 points *below* the authors' code, so the gains below are, if anything, understated.

6.3 Main result: every method under every readout (K16, long tail)

GeFL-F, GeFL-F + LA, the minimal variant and our method, trained once per seed (IF = 100, Dirichlet 0.5, K = 10, 3 seeds), each scored with its heads as trained and under every readout (tables T19 / T20; balanced accuracy).

	MNIST	FMNIST	SVHN	CIFAR-10	CIFAR-100
GeFL-F, as trained	74.5	59.2	49.6	33.9	7.8
GeFL-F + ER	90.4	79.3	64.7	51.3	15.2
GeFL-F + CCVR (all clients)	91.9	80.5	64.8	49.9	14.3
minimal variant + ER	92.4	80.6	66.4	53.4	16.5
Ours, as trained	95.1	80.1	65.4	45.7	15.4
Ours + ER	**94.4**	**81.5**	**66.7**	**53.9**	**18.6**
Ours + CCVR (all clients)	95.5	82.6	67.2	52.6	17.8
Ours − GeFL-F, both + ER [95% CI]	**+3.9** [3.0, 4.8]	**+2.2** [0.1, 4.4]	**+2.0** [0.8, 3.2]	**+2.6** [2.1, 3.1]	**+3.4** [3.1, 3.8]
Ours − GeFL-F, both + CCVR (all clients)	+3.6	+2.1	+2.4	+2.8	+3.5
Ours − GeFL-F, both + CCVR (own clients)	+16.0	+13.9	+2.2 (n.s.)	+2.5	+4.0
oracle (pooled real refit), Ours − GeFL-F	+4.1	+2.6	+4.0	+3.4	+4.5

- **The readout is where most of the long-tail loss sits.** ER lifts GeFL-F by 15.9 / 20.1 / 15.1 / 17.4 / 7.4 points. Compared with un-recalibrated GeFL-F our method gains 8–21 points; against a recalibrated GeFL-F the margin is 2.0–3.9 with ER and 2.1–3.6 with CCVR. These are the margins we claim. They pass the threshold we fixed in advance (≥ 2 points, interval excluding zero) on four of the five datasets; SVHN is at +1.98.
- **The representation gain is consistent.** Under ER, CCVR and the oracle our bodies support a better last layer on all five datasets (+2.0 to +4.5); under the weaker Fed3R and FedNCM readouts the margin is +1.1 to +6.1, smallest on SVHN, CIFAR-10 and CIFAR-100 under Fed3R. The anchored generator, not the readout, is what differs.
- **The readout matters for our method too, but less.** Our heads as trained are close to their best readout on MNIST, SVHN and CIFAR-100 but not on CIFAR-10 (45.7 as trained, 53.9 with ER), where GeFL-F + ER (51.3) beats our heads as trained. Hence the recipe computes ER for every method, ours included.
- **Published per-architecture protocols are weak here.** With K = 10 each architecture has one client, so CCVR, Fed3R and FedNCM restricted to an architecture's own clients see one client's data; they lose 4–26 points against their all-client versions. Our margin is largest there (+14 to +16 on MNIST / FashionMNIST) but we do not headline it.

- **Further metrics** (heads as trained, ours vs GeFL-F): worst-class recall 87.2 vs 28.6 (MNIST), 32.6 vs 18.9 (FashionMNIST), 31.5 vs 6.4 (SVHN), 26.6 vs 5.8 (CIFAR-10); ECE 1.1 vs 8.3, 11.2 vs 23.5, 7.3 vs 23.9, 31.0 vs 48.0, 35.8 vs 45.7. Every readout worsens calibration.
- **CIFAR-100.** Every method is weak in absolute terms (15–19% with a readout): the frozen 10-channel extractor is the bottleneck. The margin (+3.4, the tightest interval of all) shows the effect is not specific to ten classes.

6.4 Fixing the collapse: optimiser vs aggregation (K15)

See §3.4: removing coupled decay recovers 2–11 points; HWA adds 1.3–1.8 at K = 50 / 100 on MNIST.

6.5 GeFL-F's balanced setting (K16, IID)

balanced accuracy	MNIST	FMNIST	CIFAR-10	SVHN	CIFAR-100
GeFL-F, as trained / + ER	95.7 / 96.9	83.0 / 84.5	56.8 / 61.4	73.8 / 76.2	12.5 / 19.8
minimal variant, as trained / + ER	96.7 / 97.1	83.5 / 84.5	58.6 / 62.1	74.7 / 76.4	14.5 / 20.1
Ours, as trained / + ER	97.4 / 97.4	84.0 / 84.8	57.8 / 61.6	76.0 / 77.3	18.0 / 21.4
Ours − GeFL-F, as trained / + ER	+1.7 / +0.6	+0.9 / +0.3 (n.s.)	+1.0 / +0.1 (n.s.)	+2.2 / +1.1	+5.5 / +1.6
`best_mean_acc` GeFL-F / Ours (paper GeFL-F)	96.0 / 97.4 (95.47)	83.2 / 84.1 (83.14)	59.3 / 60.0 (55.86)	74.9 / 77.2 (76.26)	12.9 / 18.0 (–)

No harm in the balanced setting, and a small gain on the harder datasets: with ER the difference is +0.1 to +0.6 on MNIST / FashionMNIST / CIFAR-10 and +1.1 (SVHN, p = 0.03) / +1.6 (CIFAR-100, p = 0.01); on CIFAR-10 the minimal variant is slightly ahead of ours. Paper values are single published numbers; we make no claim against the paper's other nine methods.

6.6 Fair re-test on new seeds, settings chosen on validation (K17)

All numbers above use seeds 0–2 and settings chosen on the test sets. K17 repeats the comparison on new seeds 3–5. For every method, its readout (heads as trained, BBC, ER, Fed3R, FedNCM, CCVR; all-client or own-client) is chosen on the held-out validation set; for ours, $T_s \in \{1, 10\}$ is chosen the same way. The selection rule, four predictions and the analysis (`experiments/k17_analysis.py`, table T22) were committed before any run. K19 adds three published baselines on the same splits and seeds: the Gaussian generator built from exact statistics with LA (FSG + LA, CCVR-like virtual features used during training), and server-side ensemble distillation of every architecture's head into each head on generated features (FedDF), with and without LA. Test-set balanced accuracy, ours minus baseline, paired over seeds, 95% CI, Holm-corrected over the nine baselines:

	MNIST	FMNIST	SVHN	CIFAR-10	CIFAR-100
Ours (selected T_s, readout)	95.6 (10, CCVR)	82.1 (10, CCVR)	65.6 (10, CCVR)	53.1 (1, ER)	18.7 (10, ER)
− GeFL-F	**+3.4** [3.2, 3.6]	+1.0 [−0.0, 2.1]	+1.5 [0.6, 2.5]	**+2.4** [1.8, 3.0]	**+4.0** [2.6, 5.4]
− GeFL-F + AdamW + LA (decay fix)	**+2.0** [1.9, 2.2]	+0.5 [−0.2, 1.2]	+1.5 [0.4, 2.6]	+1.5 [0.8, 2.2]	**+3.0** [2.3, 3.7]
− real-data heads + LA (balanced softmax)	**+4.7**	**+2.6**	**+11.3**	+1.5 (n.s.)	**+1.6** [0.7, 2.6]
− real-data heads + FedLC / FedRS	**+4.7**	**+2.6**	**+11.2**	+1.6 (n.s.)	**+1.7 / +1.8**
− FSG + LA	**+2.9** [1.7, 4.1]	**+1.3** [1.3, 1.4]	**+12.4**	+0.5 [−1.1, 2.1]	**+1.6** [0.4, 2.8]
− FedDF	**+1.9** [1.4, 2.5]	+0.3 [−0.6, 1.2]	+0.5 [−1.0, 2.0]	+2.3 [0.9, 3.7]	**+3.5** [2.5, 4.5]
− FedDF + LA	**+2.0** [1.2, 2.8]	+0.3 [−0.4, 1.0]	−0.3 [−1.6, 1.0]	+1.7 [0.4, 3.0]	**+3.5** [2.2, 4.9]
margin over the strongest baseline	**+1.9** (FedDF)	+0.3 (FedDF)	−0.3 (FedDF + LA)	+0.5 (FSG + LA)	**+1.6** (FSG + LA)

Bold: Holm-adjusted p < 0.05 (over nine baselines; with nine instead of six, the decay-fix rows on SVHN and CIFAR-10 lose significance).

- **Ours is not ahead of every baseline.** Against the strongest one — FedDF on MNIST, FashionMNIST and SVHN, FSG + LA on CIFAR-10 and CIFAR-100, each with its own validation-selected readout — the margin is −0.3 to +1.9 points, significant on MNIST and CIFAR-100 only; FedDF + LA is 0.3 points ahead on SVHN (n.s.). The pre-registered P1 (≥ 2 points over every baseline on ≥ 4 datasets) **fails**, and so does K19's own prediction (≥ 2 points over each of the three on ≥ 4 datasets: it holds on 2).
- **FedDF is strong once it gets an exact readout.** Server-side distillation of the ensemble on GeFL-F's own synthetic features (collapsed generator included) improves the bodies enough that, with ER or CCVR, it matches the decay fix or better. This is consistent with §4.3: distillation targets carry the ensemble's information even where the generator's labels carry none.
- **Where the gain comes from.** Removing coupled decay plus an exact readout recovers most of GeFL-F's long-tail loss; the anchored generator adds 0.5–3.0 on top (significant on MNIST and CIFAR-100). On CIFAR-100 GeFL-F's own generator is *harmful* (14.8 vs 16.9 for no synthetic data at all, each with its validation-selected readout); ours turns it into a gain (+1.6 over no synthetic data). §4.3 explains the sign: GeFL-F's synthetic features on CIFAR-100 carry almost no label information ($s = 0.002$), and label-independent synthetic data penalise the spread of the logits (Proposition 9).
- **The four real-data baselines are one baseline.** LA, FedLC, FedRS and balanced softmax give the same body (Proposition 7), hence the near-identical rows above; we keep them because reviewers ask for them by name.
- **P2 fails** (3 of 5): validation picks $T_s = 10$ on SVHN and CIFAR-100, where the earlier test-chosen rule said 1.

- **P3 holds:** on CIFAR-10, near the gate threshold, the anchor branch and the forced class-row branch are within 0.13 points.
- **P4 fails as pre-registered** (Spearman 0.28 pooled over datasets): the label-free head–ER disagreement tracks the readout gap within each dataset (0.82–0.97, exploratory) but not across datasets of different difficulty.
- **Reproducibility.** MNIST was run twice (two sessions, same seeds): the 21 paired runs differ by -0.06 ± 0.43 points with ER.
- **Ablation, five seeds (K09 + K14, complete).** As trained, removing any component lowers balanced accuracy on MNIST in every one of the 5 seeds (LA −3.05, PC −3.89, CSL −1.35, MC −1.13, KH −0.67); on FashionMNIST LA −5.78 and CSL −1.50 (5 of 5), KH −0.65 (5 of 5), MC −0.63 and PC −0.50 (4 of 5). With ER (seeds 3–4): LA +0.11 / +0.05 (its contribution vanishes, as Proposition 7 says it must), PC −1.05 / +0.15, CSL −0.44 / −0.53, MC −0.53 / −0.14, KH −0.26 / −0.13 (MNIST / FashionMNIST). The covariance variant MC-B is +0.75 / +1.71 as trained and +0.43 / +0.56 with ER (seeds 3–4).
- **Mechanism controls (K20; one generator per seed shared by every arm, FashionMNIST and CIFAR-10, seeds 3–4).** All three pre-registered predictions hold. M1: herding beats a uniformly random subset of the same $4n$ candidates (FashionMNIST −0.49 as trained / −0.16 ER when replaced; CIFAR-10 −0.23 / −0.23), so KH's effect is the selection, small but consistent. M2: MC's gain comes from the spread: spread-only matches full MC (+0.29 / +0.15 as trained), mean-only loses 0.9–1.0 points as trained (0.4 with ER) although it *raises* the referee's tail fidelity by 4–5 points — recognisable samples are not the same as useful samples. M3: results are flat in the RFF dimension (1,536 vs 6,144) and the pool size ($8n$ vs $4n$), within 0.2 points.

6.7 Where the remaining loss is: comparison with centralised training

We train the centralised comparator of §2.3 on the same long-tailed splits (N7, CPU, `docs/research/math_checks/`): the extractor either from GeFL-F's federated Stage I or trained centrally (class-balanced or instance sampling), then each architecture's body on the **pooled** real features with the Stage-III schedule, then ER. Balanced accuracy, mean over the 10 architectures; MNIST and FashionMNIST seed 0 (seed 1 as a check), CIFAR-100 seed 0 on architectures 0 / 4 / 8.

Bodies (federated extractor, ER readout).

	MNIST	FashionMNIST
one client's real data only	91.5	78.1
ours (K16, same split)	94.4	81.1
pooled real data, instance sampling	**94.4**	**81.9**
pooled real data, class-balanced sampling	94.2	81.5

- A body trained on one client loses 3.0 / 3.8 points against the pooled body (seed 1: 3.7 / 4.2). This is the federated cost of having one client per architecture.
- Our synthetic data recover all of it on MNIST and 80% on FashionMNIST on seed 0, and all of it on both on seed 1 (94.2 vs 94.1; 81.3 vs 80.9).
- On CIFAR-100 the one-client body reaches 15.1 and the pooled body 18.1 (three architectures); ours + ER is 18.7 (ten architectures, K16), so it is at the pooled-body level, approximately (± 1).
- **[PENDING: P0]** SVHN and CIFAR-10, where real-only bodies are weakest and s is moderate, are not yet measured.

Extractor. Replacing the federated extractor by a centrally trained one:

pooled body + ER	MNIST	FashionMNIST	CIFAR-100
federated extractor	94.4	81.9	18.1
centralised, class-balanced	94.4	83.0	17.1
centralised, instance sampling	93.9	82.3	–

On raw extractor features the federated extractor is less linearly separable (balanced LDA 88.6 vs 90.4 on MNIST, 85.8 vs 89.6 on seed 1, 78.8 vs 79.9 on FashionMNIST), but a trained body recovers the difference on MNIST (94.1 vs 93.9 on seed 1). On FashionMNIST about 1 point remains, most of it the long-tailed weighting of the extractor's loss (0.7 against a centralised ERM extractor) rather than federation (0.4), as §2.4 predicts. On CIFAR-100 the centralised balanced extractor is 1.0 *worse*; it had 30 pooled epochs against Stage I's 50 rounds of 5 local epochs. Exact recalibration of the averaged BatchNorm statistics changes accuracy by ≤ 0.3.

A federated fix for the extractor's share (X4 pilot, P2 of the analysis). Weighting each sample's Stage-I loss by $N/(Cn_y)$ from the exact global counts (clipped at 20; one released count vector) makes the federated objective, to first order, the balanced one (Corollary 2). On the validation split (2,000 held-out images; seeds 0 / 1), against the plain federated extractor: raw-feature LDA +1.2 / +2.3 (MNIST) and +0.8 / +0.2 (FashionMNIST); one-client bodies + ER +0.5 / +0.3 and +0.1 / +1.9; pooled bodies + ER +0.2 / +0.0 and +1.0 / +1.1. The predicted +1 ± 1 on FashionMNIST and 0 ± 0.5 on MNIST held. **The full pipeline did not confirm it (K21, ours + BALFE, seeds 3–5, ER):** +0.19 (MNIST), +0.01 (FashionMNIST), −8.2 (CIFAR-10) and −4.2 (CIFAR-100; one of the three seeds diverged to chance). The weights are not normalised within a batch, so on clients rich in rare classes they act as a step size up to 20 times larger; on the CIFAR extractor (390 parameters, 50 rounds) this destabilises Stage I. A normalised or sampling-based version was not run.

Decomposition.

	MNIST	FashionMNIST	CIFAR-100
ours + ER, balanced data (K16)	97.4	84.4	21.4
ours + ER, long tail (K16)	94.4	81.1	18.7
long-tail drop of our federated pipeline	**2.9**	**3.3**	**2.7**
centralised, pooled body, balanced data	97.6	85.7	22.2
centralised, pooled body, long tail, best extractor	94.4	83.0	18.1
long-tail cost of the centralised pipeline	**3.2**	**2.7**	**4.1**
federated residual, long tail (centralised − ours)	0.0	1.9 (≈ 1.1 extractor + 0.8 body)	≈ −0.6
federated residual, balanced data	0.3	1.3	0.8

- On these three datasets our federated pipeline loses about as much to the long tail as the centralised pipeline does. The largest remaining loss is the long-tailed data itself, not federation.
- What federation still costs is ≈ 0 on MNIST, ≈ 1.9 on FashionMNIST (of which ≈ 1.3 is present on balanced data too) and nothing measurable on CIFAR-100.
- **Ranking of the losses for GeFL-F** (federated vs centralised, same pooled long-tailed data, same frozen-extractor design): last layer 7–20 points (fixed exactly by ER, Theorem

10); generator-row collapse 7–11 at $K = 10$ (fixed by the decay fix, HWA or the mean anchor); one-client bodies 3.0–4.2 (closed by high-s synthetic data); synthetic data with $s \approx 0$ up to -2.2 (removed by raising s); row dilution 1.3–1.8 at $K = 50/100$ (HWA); federated extractor 0–1.1 (not touched by any of our changes).

- *Why the margin over strong baselines is small (§6.6).* The decay fix removes the generator term and ER the readout term; what is left for a better generator is the body term, part of which the decay-fixed GeFL-F's generator already recovers (+2.6 / +2.8 over real-only bodies on MNIST / FashionMNIST, against +4.7 / +3.4 for ours). If the pooled body is the ceiling — a conjecture, since synthetic data can also act as augmentation — our margin over the decay fix is bounded by $94.4 - 92.5 = 1.9$ on MNIST, where we measure +2.0, and by $81.9 - 80.5 = 1.4$ on FashionMNIST, where we measure +0.5. These compare one N7 seed with three K17 seeds and are indicative only.

6.8 What did not work

- **Exact readout during training (ERL, pilot).** Recomputing the balanced LDA last layer every round (or every 5) and training only the bodies under it gave 93.9 / 79.4 on MNIST / FashionMNIST against 94.1 / 79.8 for ER computed once afterwards (reduced schedule, one seed); for GeFL-F 90.0 / 78.4 against 91.5 / 78.9. A pooled-real refit of the ERL bodies is lower (MNIST 89.2 vs 94.9): training under a fixed balanced readout makes the representation slightly worse, and calibration much worse (ECE 41% on FashionMNIST). Dropped by its pre-registered rule.
- **Signal-fraction gate for our method (K21, §4.3).** Helps the decay-fixed GeFL-F on the CIFARs but not ours: the LDA-based estimate of s_c saturates under moment correction. A better estimator (e.g. a QDA or a small network fitted from released second moments) is open.
- **Class-balanced Stage-I loss (BALFE, K21, §6.7).** Positive in the CPU pilot, neutral on MNIST / FashionMNIST and harmful on the CIFARs in the full pipeline (unnormalised weights).
- Earlier negative results (prior-completing batches, lazy decay on top of HWA, the Gaussian and hybrid generators, server-side distillation, Bayesian consensus labels, real–synthetic alignment, VCT vs MixUp) are in Appendix B.7 and B.11.

6.9 Privacy and cost

Formal DP on the four added statistics (E26): 4.5–6.9 points at $\varepsilon = 8$ and 9.8–14.1 at $\varepsilon = 2$, still above GeFL-F at $\varepsilon = 2$ on MNIST (+9.8) and FashionMNIST (+7.1, p = 0.084); details and scope in Appendix B.6. Cost: Ours is 2.2–2.8× slower than GeFL-F end to end (kernel herding is 83–88% of the head stage); ER adds one round and up to about 5×10^5 numbers per architecture per client (§5.6); details in Appendix B.12.

Membership inference on the released class sums (X14, one seed, `docs/research/math_checks/x14_membership.py`). An attacker who knows a target's label and holds public reference data scores it by how far the released class mean has moved toward it. Members vs non-members, AUC:

release	MNIST all / tail / rarest class	FashionMNIST all / tail / rarest class
exact sums	0.52 / 0.64 / 0.69	0.52 / 0.63 / 0.73
Gaussian mechanism, $\varepsilon = 8$	0.50 / 0.58 / 0.64	0.52 / 0.54 / 0.45
Gaussian mechanism, $\varepsilon = 2$	0.50 / 0.53 / 0.61	0.51 / 0.50 / 0.41

Head classes are at chance (0.49–0.51); the exact release leaks membership for rare classes

(24–112 samples), as expected for a mean over few samples, and the noise of E26 removes most of it. The rarest-class AUCs rest on 24 members and are noisy (FashionMNIST at $\varepsilon = 8$ / 2 is below 0.5). With $\varepsilon = 2$ the DP guarantee itself allows an AUC up to $e^{\varepsilon}/(1 + e^{\varepsilon}) \approx 0.88$, so these values are consistent with it. Secure aggregation alone does not protect rare classes; the added statistics should be released with noise when some class is rare.

6.10 More clients, partial participation and other imbalance settings (K18, K22, K23, K26)

Several clients per architecture (K18, SVHN long tail, seeds 0–2). ER / heads as trained:

	K = 50	K = 100
GeFL-F	58.0 / 42.2	58.8 / 42.2
GeFL-F + AdamW + LA (decay fix)	60.3 / 54.3	59.9 / 51.4
GeFL-F + HWA + LA	60.3 / 54.8	60.8 / 53.2
Ours	**61.8 / 58.2**	**61.0 / 55.2**

Pre-registered: P1 (HWA − decay fix larger at K = 100 than at K = 10) holds (+1.8 vs ≈ 0 as trained; +0.9 vs −0.1 with ER); P2 (ours + ER ≥ GeFL-F + ER + 2 at both K) holds (+3.8 / +2.2); P3 (own-client CCVR within 2 points of all-client CCVR at K = 100, where each architecture has 10 clients) **fails**: own-client CCVR is still 12.6–15.2 points lower for every method, so the published per-architecture readouts stay weak even with ten clients per architecture.

Partial participation (K23, K = 100, 10% / 20% of the clients per round, the same seeded subset for every method; 31 of 48 runs so far). On FashionMNIST at 10% participation HWA beats the decay fix by +3.6 / +2.7 points as trained (two seeds) but not with ER (−0.04 / −0.11); ours reaches 73.2 as trained and 79.8 with ER against 69.1 / 78.4 for HWA. **[PENDING: NB71]** The MNIST decay-fix and ours arms and the remaining FashionMNIST seed.

CIFAR-10 with five clients per architecture (K26, K = 50, long tail, seeds 3–5). Test balanced accuracy:

	as trained	+ ER	tail classes, as trained
GeFL-F + AdamW + LA (decay fix)	35.7	45.1	21.4
GeFL-F + HWA + LA	35.9	45.2	22.5
Ours (gated; anchor branch on every seed, F = 0.08–0.10)	**40.4**	**46.1**	**31.2**

Pre-registered: D1 (HWA − decay fix ≥ 0) holds, but only just: +0.24 [0.08, 0.40] as trained and +0.10 [−0.22, 0.41] with ER, far from MNIST's +1.8 at K = 50 (K15). D2 (ours + ER ≥ both baselines + ER) holds: +1.0 [0.2, 1.8] over the decay fix and +0.9 [−0.0, 1.9] over HWA; as trained the margin is +4.7, and +9.8 on the tail classes. Every method is 6.5–7 points below its K = 10 value on the same seeds (K17: 51.6 and 53.1 with ER), so splitting each architecture's data over five clients costs more than any method recovers.

Other imbalance settings and random rare classes (K22, seeds 3–5, ER). FashionMNIST / SVHN:

	IF = 10	Dirichlet 0.1	Dirichlet 1.0	random rare classes
real-data heads + LA	80.8 / 62.8	76.6 / 51.1	78.4 / 56.9	77.7 / 54.9
decay fix (AdamW + LA)	83.2 / 72.3	78.9 / 62.1	80.8 / 65.2	79.5 / 63.7
Ours	**84.2 / 74.8**	**80.4 / 63.9**	**81.5 / 66.2**	**80.1 / 64.8**
ours − decay fix, SVHN [95% CI]	+2.6 [1.0, 4.1]	+1.7 [1.0, 2.5]	+1.0 [−0.0, 2.0]	+1.1 [0.8, 1.3]

Pre-registered: B1 (ours ≥ the strongest of the two baselines in all 8 cells) holds, with ER and with the validation-selected readouts; B2 (larger margin over the decay fix at Dirichlet 0.1 than at 1.0) holds on both datasets (FashionMNIST +1.5 vs +0.8; SVHN +1.7 vs +1.0); B3 (random rare classes within 1 point of the K17 margin) holds on both (FashionMNIST +0.5 vs +0.5; SVHN +0.9 vs +1.5 with the validation-selected readouts, +1.1 with ER). The classes are relabelled by a seeded permutation before the split, so which classes are rare changes per seed. The margins are small (three seeds, not Holm-corrected; the Dirichlet 1.0 interval on SVHN touches 0), so this is evidence that the ordering is not specific to one imbalance setting, not a new effect size. On SVHN the real-data heads are 7–11 points behind both synthetic-data methods in every setting, as in K17.

6.11 Beyond GeFL: rare-item collapse in federated recommendation (E40)

The mechanism of §3 needs only a row of parameters whose data gradient is non-zero only on samples that use it, clients that never use it, local Adam with coupled L2, and plain averaging. Federated matrix factorisation has all of them (FedMF-style: item embeddings and biases shared and averaged, user embeddings local). MovieLens-1M explicit ratings, 90 / 10 split per user, users assigned at random to K = 10 or 50 clients, $d = 32$, 30 rounds of one local epoch, Adam lr 10^{-3}, $\lambda = 10^{-4}$, 3 seeds; predictions E1–E4 written in `experiments/fedembed.py` before the runs. Items with $\kappa_c < 1$ (895 of 3,706 at K = 10; 2,108 at K = 50):

arm	collapsed (≤ 0.15 of initial norm), K = 10 / 50	RMSE on their test ratings, K = 10 / 50	RMSE overall, K = 10 / 50
Adam + coupled L2, FedAvg	85.0% / 83.4%	1.094 / 1.064	0.859 / 0.908
same, optimiser state reset every round	99.7% / 100%	1.107 / 1.109	0.882 / 0.950
AdamW (decoupled), FedAvg	0% / 0%	1.036 / 0.992	0.904 / 0.907
Adam, no decay	0% / 0%	1.036 / 0.992	0.904 / 0.907
Adam + L2, rows averaged over holders (HWA)	23.3% / 8.7%	1.042 / **0.967**	**0.857** / **0.894**
SGD + L2, FedAvg	0% / 0%	1.160 / 1.144	0.967 / 1.010
centralised Adam + coupled L2	**78.0%**	1.099	0.888

- **The count rule ranks the collapse** (Spearman(log κ_c, final norm) 0.93 / 0.96), and the collapse grows with K (22% → 48% of all items; E4 holds). The accuracy cost is real: Adam + L2 is worse than AdamW and HWA on these items at both K (E3 holds).

- **Two predictions fail.** E1 ("every $\kappa_c < 1$ row collapses"): 83–85% do; the rest end at 0.15–0.3 of their norm, the items with the most ratings among them. E2 ("no collapse centrally"): centralised Adam + L2 collapses 78% of the same items.
- **One rule covers both worlds.** Adam's second moment forgets with time constant $1/(1-\beta_2) = 1{,}000$ steps; a row that receives data less often than that gets near-full-size decay steps between its data steps, centrally too. The share of all optimiser steps whose batch carries data for the item predicts the final norm centrally (Spearman 0.91) and federated (0.94–0.97); items seen less often than about once per 500 steps are wiped out centrally. At K = 50, federation collapses moderately rare items (seen every 100–1,000 steps) about 10 times harder than centralised training (median norm 0.006–0.016 vs 0.07–0.13; `docs/research/math_checks/x5_step_frequency_mf.py`).
- **Practical reading.** Coupled L2 regularises popular items well (best overall error at K = 10) and destroys rare ones; AdamW removes both effects; HWA keeps the regularisation and gives the best rare-item error at K = 50.
- *Novelty.* Count-weighted item aggregation is known (FedNCF [66]) and weak or skewed item embeddings in federated recommendation have been reported [67]; we found no account of the optimiser mechanism or of a count-based predictor.

7. Related work

Model-heterogeneous FL by distillation and partial sharing. FedAvg [4] assumes that every client trains the same architecture, so that parameters can be averaged. A first family of methods replaces parameter averaging with an exchange of predictions. FedMD [5] and DS-FL [8] share client outputs on a public or unlabelled dataset and distil the consensus back into each model. FedDF [6] fuses heterogeneous client models at the server by ensemble distillation on unlabelled or generated inputs. FedGKT [7] moves a large model to the server and trains it by distilling from small edge models. AvgKD [9] analyses these protocols in a kernel-regression model and shows that distillation-based fusion degrades under data heterogeneity. A second family shares only part of the network. LG-FedAvg [10] keeps local representations and averages a global head. FedRep [11] does the reverse: it learns a shared representation and personal heads. FedProto [12] exchanges class prototypes instead of parameters. GeFL-F [3] combines both ideas: it shares a frozen feature extractor and a federated feature generator, and it leaves the heads architecture-specific. Our work keeps GeFL-F's communication pattern and architecture freedom unchanged. It studies what that pattern does to rare classes when the shared component is a FedAvg-trained *generator*.

Generator-based and data-free knowledge transfer. FedGen [13] trains a lightweight generator in feature space at the server. The generator is fitted so that the ensemble of client predictors classifies its samples as the conditioning label, and clients then regularise local training with these samples under that hard label. Its supplementary variant weights each client's predictor per label by that client's label count. FedFTG [14] uses a server-side generator to fine-tune the global model by data-free distillation. It samples labels in proportion to global class counts and weights each client's teacher loss per class by the client's share of that class. DENSE [15] trains a generator against the client ensemble and distils the ensemble into a global model in one round. FedCG [16] has clients share conditional GANs in place of their private extractors. FedCVAE [17] fits client conditional VAEs for one-shot FL under extreme heterogeneity. Felo/Velo [18] trains a server-side conditional VAE on clients' mid-level features and sends synthetic features and per-class averaged logits back to heterogeneous clients. FedVTC [19] shares feature mean and covariance statistics, from which clients generate data to fine-tune heterogeneous models; it is the closest precedent for a generator driven by federated class statistics. GeFL [3] trains the generator itself by FedAvg across clients, with CVAE, DCGAN or DDPM backbones, in image space (GeFL) or feature space (GeFL-F). FedGen and FedFTG apply label-count weights to the *teacher predictions or losses* that train a server-side generator. HWA instead weights

the *parameters* of a client-trained generator, namely its per-class conditioning rows, during aggregation. It is derived from a failure that is specific to that setting: under FedAvg with Adam's coupled weight decay, non-holders shrink a class's row at Adam's step size while holders barely grow it, so tail rows contract even when most clients hold the class. Closely related ideas exist elsewhere: federated recommendation aggregates each item embedding over the clients that updated it (FedNCF, Perifanis & Efraimidis, Knowledge-Based Systems 2022), and class embeddings can collapse under FedAvg when clients see only positive labels (FedAwS, Yu et al., ICML 2020), a different cause from decay. Class-conditional collapse in centralised generators is also known (Shahbazi et al., ICLR 2022; Qin et al., CVPR 2023; Rangwani et al., ECCV 2022). Per-class aggregation over the clients that hold a class also appears in cwFedAvg [31] and in FedProto's prototype averaging [12]; HWA applies the same rule to a generator's class rows. What we add is the federated mechanism: coupled Adam decay on non-holders combined with flat averaging.

FL under class imbalance and long tails. Much of this work locates the bias in the classifier. CCVR [20] re-calibrates the classifier after training on virtual features sampled from a Gaussian mixture built from federated feature statistics. Our federated Gaussian sufficient-statistics generator is close to this idea. CReFF [21] re-trains the classifier on "federated features" optimised so that their gradients match those of real data. FEDIC [22] uses ensemble distillation with logit adjustment and a calibration gating network. A second group changes the local loss. Wang et al. [24] infer each round's class composition with a monitor and counter it with a Ratio Loss. FedLC [23] calibrates logits by each client's label frequencies through pairwise margins. FedRS [29] restricts softmax updates for classes a client does not hold, and MAP [30] develops this restricted-softmax view for incomplete classes. BalanceFL [26] changes local training so that the uploaded model behaves as if trained on balanced data. CLIMB [25] casts class imbalance as a constrained learning problem. Fed-GraB [27] re-weights gradients using an estimate of the global long-tailed prior, and FedLoGe [28] combines a shared backbone with a static ETF classifier and per-client classifiers. Class-aware aggregation also appears for classifier layers: cwFedAvg [31] averages per class with weights given by client class distributions, and FedLA [32] weights client updates using both data size and label distribution. Our LA component is a direct application of logit adjustment by the client's own prior. It is closely related to FedLC and to the logit adjustment in FEDIC, and we do not claim it as new on its own; FedRoD [51] trains a balanced-softmax head with client counts, which is the same correction. What is new is the setting, a generator and heads trained in separate phases, and the finding that LA helps only after the generator is repaired. Post-hoc recalibration from federated statistics is established: CCVR [20], Fed3R [52] (closed-form ridge classifiers from federated sums) and FedNCM [53] (nearest-class-mean heads) all compute a classifier from aggregated class statistics, and FedNPC [54] calibrates federated long-tailed classifiers after training. These are the natural baselines for the readout gap, and every one of them is run on every trained model (§6.3, §6.6). The principle "take class information from exactly aggregated sums" is therefore not new; our use of it is to anchor and correct a generator.

Long-tailed recognition. The class-balanced loss [33] re-weights classes by the effective number of samples $E(n) = (1-\beta^n)/(1-\beta)$, a saturating model of how much new information extra samples add. LDAM [35] enforces label-dependent margins and defers re-weighting. Decoupling [36] shows that instance-balanced representations with a re-balanced classifier are hard to beat. Logit adjustment [34] adds $\tau \log \pi$ to the logits during training (or subtracts it post hoc), which is Fisher-consistent for the balanced error. We borrow the effective number as an inverse-variance weight for *holders* of a class, not as a loss weight. We use logit adjustment with a per-client rather than a global prior, which needs no communication. Our diagnosis also depends on the difference between coupled and decoupled weight decay in Adam [45].

Synthetic data, label quality and soft labels. Knowledge distillation [37] trains a student on a teacher's soft predictions, and label smoothing [38] mixes the one-hot target with a

uniform distribution, which can remove inter-class information useful to a student [39]. Menon et al. [40] show that a teacher that approximates Bayes class-probabilities lowers the variance of the student's objective, with a bias–variance trade-off when the approximation is poor. Work on training classifiers with generated data finds that label fidelity of samples, not only visual quality, limits accuracy. The Classification Accuracy Score [41] trains on generated samples and shows large drops relative to real data, with class-specific failures. Later studies of diffusion-generated training data report gains from synthetic data together with clear limits [42, 43, 44]. CSL applies this view to generated *features* in heterogeneous FL. Its target mixes the one-hot label with the mean softmax of all architectures' heads, so it is label smoothing toward a learned ensemble posterior rather than toward a uniform distribution. The bias–variance argument of [40] explains why the mixture ($\beta = 0.5$) beats the pure ensemble ($\beta = 1$), and why lower-bias labels make a larger synthetic budget useful. Unlike FedDF, DENSE or our negative server-side variant, CSL does not distil at the server. It only relabels the clients' synthetic phase. Feature-space generators for classifiers are established outside FL (f-CLSWGAN [55], Delta-encoder [56]), and a generator trained by MMD against random-feature class embeddings is DP-MERF [57], which our server-only KME-Gen reproduces on federated sums.

Privacy. Secure aggregation [46] lets a server learn only the sum of client vectors. HWA needs only the sums $\sum_k E(n_{k,c})\, w_c^{(k)}$ and $\sum_k E(n_{k,c})$, so it runs inside such a protocol and reveals neither individual rows nor individual histograms. Differential privacy [47] and its use in deep learning [48] and in federated training at the user or client level [49, 50] give formal guarantees. Our DP experiments add Laplace noise to class counts (minimal variant) and Gaussian noise to the four statistics the anchored method adds (E26). Neither covers the generator's FedAvg training or the heads; a formal accounting for those is left to future work. Secure aggregation is simulated in our code: the server is handed the sum.

Federated optimisation theory. FedNova [58] shows that FedAvg with heterogeneous local work converges to a stationary point of a surrogate objective whose client weights are $p_k a u_k$; Charles and Konečný [59] and Pathak and Wainwright [60] characterise the fixed points of local-update methods, which in general are not stationary points of the pooled loss, and SCAFFOLD [61] removes the client drift with control variates. We use these facts only in their simplest exact forms (Corollary 2; Propositions 6 and 8 are fixed-point formulas for quadratic models), to say which stage of GeFL-F is consistent with pooled training and which is not. Minority collapse [62] describes the last-layer geometry of imbalanced centralised training; a head trained on one client is its extreme case, which ER sidesteps. Training bodies under a fixed head (FedBABU [64]) is related to our negative ERL pilot (§6.8).

7.7 Positioning relative to the closest works

GeFL/GeFL-F [3] was evaluated only with IID clients and aggregates every generator parameter by flat FedAvg. We show that, combined with Adam's coupled weight decay, this contracts the conditioning rows of tail classes even when most clients hold them (Theorem 3), and HWA removes exactly the non-holder term while reducing to FedAvg when class counts are equal. FedGen [13] and FedFTG [14] also use label-count weights, but on the teacher predictions or losses that train a *server-side* generator; we weight the *parameters* of a *client-trained* generator. Both label generated samples with the hard conditioning label, whereas CSL relabels them with a one-hot/cross-architecture-consensus mixture. CReFF [21] and CCVR [20] use synthetic or virtual features to re-train one homogeneous classifier after training, while we keep per-architecture heads and change how the generator is aggregated and its samples labelled throughout training. FedLC [23] already debiases local training with client label frequencies; our LA belongs to the same family, and our claim for it is only empirical: it is needed, but it pays off fully only after HWA. FedDF-style server distillation [6] of the ensemble on generated features lowers IID accuracy in our hands. CSL uses the same ensemble only as a low-variance label inside the clients' synthetic phase, and this is what makes a larger synthetic budget pay off. The closest

labelling scheme is Felo/Velo [18]. Its server averages clients' logits *per class* (one vector per class, computed on real data) and uses them as KL targets on clients' *real* samples; its generated features serve only as MSE targets for the feature extractor. CSL instead labels *each generated feature* with the consensus of all architectures' heads on that very feature, and trains the heads on it. The per-sample consensus is what tracks the generator's per-sample label error, the bias b of Proposition 11.

8. Limitations

- **Margins.** On new seeds with validation-chosen settings, our margin over the strongest of nine baselines is -0.3 to $+1.9$ points (significant on MNIST and CIFAR-100 only; FedDF + LA is 0.3 ahead on SVHN, n.s.); the pre-registered $\geq$ 2-point prediction failed. The 8–21 point gains over un-recalibrated GeFL-F are mostly a readout and optimiser effect. The paper's claim is the stage-by-stage account, not a new state of the art.
- **Readout protocol.** The strong (all-client) readouts need each architecture's body broadcast once to every client; the published per-architecture protocols are much weaker at K = 10 for every method.
- **Absolute accuracy on hard data.** With the frozen 10-channel extractor, CIFAR-100-LT accuracy is 15–19% for every method; CIFAR-10 long tail is about 54%.
- **Selection on the test set.** For seeds 0–2, T_s, β, τ, the gate threshold and the choice among variants were made on the reported test sets. K17 (new seeds, validation-selected, Holm-corrected) re-tests the main comparison; β, τ and the gate threshold were still fixed from the earlier runs.
- **HWA's contribution is modest.** Once coupled decay is removed, HWA adds 1–2 points, significant only on MNIST at 50 and 100 clients.
- **Reproducibility.** Seeds fix the splits and initialisations, but GPU training was not deterministic; a `--deterministic` option and the code commit are recorded for runs from now on.
- **Scale of the federation.** At K = 10 each architecture has one client, so per-architecture aggregation of heads is trivial there. The rarest classes are held by 6–8 of 10 clients, the rare classes are always 6–9, and CIFAR-10's realised imbalance is 48:1, not 100:1.
- **Frozen extractor and BatchNorm.** The extractor is trained once and frozen, as in GeFL-F; its BatchNorm statistics are averaged by FedAvg like any other parameter. A trainable or non-IID-robust extractor is out of scope.
- **Metric.** The paper's `best_mean_acc` is a best-over-rounds test metric. We report final-round values beside it; on CIFAR-10 IID the difference decides whether there is a gain at all.
- **Gate.** The generator gate's threshold was set after seeing SVHN. CIFAR-10 lies near it; K17 ran both branches there and they are within 0.13 points (P3), so the choice did not matter on that dataset.
- **Balanced data.** In balanced (IID) data the methods are within 0.6 points once both use a readout; on CIFAR-10 the minimal variant is slightly ahead of ours. Under label skew it is better on MNIST, FashionMNIST and CIFAR-10, but not on SVHN, where the anchor is uninformative and the gate falls back to the class rows.

- **Upper bound.** Every method is bounded by the frozen shared feature extractor, which GeFL-F's protocol fixes. On CIFAR-10 this bound (oracle ≈ 50 under the long tail) is far below the image-space GeFL numbers in the paper.
- **What the statistics reveal.** The exact statistics are federation-level sums, plus counts. Classes held by a single client are not hidden by secure aggregation; DP noise (§B.6) is the remedy, at a measured cost, and it covers only the added statistics.
- **The generator gate is set from five datasets.** F separates SVHN from MNIST and FashionMNIST by more than a factor of ten, but CIFAR-10 is within a factor of two of the threshold, and a dataset with intermediate separation would need both branches evaluated. Under the gate, the rows branch inherits HWA's cost (aggregated class counts).
- Results are on small images and on the paper's backbone sizes. Larger FEs could change the generator's failure profile.
- HWA uses aggregated class counts. LCD is the fully local alternative but is weaker.
- CSL's gain in IID data is small (+0.5 to +0.7 points), although it is consistent across seeds.
- MND in feature space is a proxy for privacy, not a guarantee. A formal DP accounting of the generator itself is future work.
- **The centralised comparison (§6.7) is narrow.** It ran on CPU with one or two seeds, on MNIST, FashionMNIST and CIFAR-100 (three of the ten architectures for CIFAR-100); SVHN and CIFAR-10 are not measured. Its extractor and bodies are fresh realisations, not the Kaggle runs' models; its one-client real-only bodies reproduce K17's within about 1 point. The centralised class-balanced extractor received fewer epochs than Stage I on CIFAR-100.
- **Two of our own fixes failed.** The signal-fraction gate (SGATE, K21) helps GeFL-F on the CIFARs but does nothing for ours: its LDA referee saturates under moment correction ($\hat{s} \approx 0.93$–1.0 for every class). The class-balanced Stage-I loss (BALFE) is flat on MNIST and FashionMNIST and loses 4–8 points on CIFAR-10 / CIFAR-100, where up to 20 times larger steps (unnormalised weights) destabilised the extractor. Both are reported (§4.3, §6.7–6.8) and were not tuned further.
- **The scale predictions are partial.** The prospective κ test (K18) ran on SVHN only and its third prediction failed; the partial-participation runs (K23) are incomplete. With five clients per architecture on CIFAR-10 (K26) HWA's advantage over the decay fix is about zero and every method loses 6.5–7 points to $K = 10$.
- **The recommendation result (§6.11) is one dataset.** MovieLens-1M, explicit ratings, 30 rounds, three seeds, one embedding size. Two of its four predictions failed: not every $\kappa_c < 1$ item collapses, and centralised Adam with coupled L2 collapses most of the same items, so in recommendation the federated part is the extra severity (about ten times smaller norms at $K = 50$), not the existence of collapse. Implicit feedback, larger catalogues and neural recommenders are not measured.
- **The theory uses models.** Propositions 6 and 8 are exact for quadratic losses; Approximation 5 neglects momentum carried between sparse steps and the drift of Adam's second moment; Corollary 2 is first order in a regime where $\eta\tau_k$ is 1–17. Each is paired with a measurement on the real pipeline, and only ranking and sign claims are drawn from them. The signal fraction s is estimated from a referee classifier trained on held-out real features.

Appendix B. Development experiments (seeds 0–2, settings chosen on test data)

These are the experiments that led to the method, kept as run. Their numbers compare against un-recalibrated GeFL-F; §B.3 gives the margins over a recalibrated baseline.

B.3 Long-tailed clients (IF = 100, Dir 0.5, K = 10; F01)

Method	MNIST bal.	MNIST tail	FMNIST bal.	FMNIST tail
FedAvg (per architecture)	62.9 ± 1.8	40.8	52.8 ± 3.0	33.7
LG-FedAvg	63.2 ± 2.2	42.0	52.5 ± 2.9	33.3
LG-FedAvg + LA	67.8 ± 1.2	50.6	57.6 ± 2.4	41.0
GeFL-F	74.6 ± 1.8	51.9	58.6 ± 3.4	35.4
GeFL-F + LA	78.9 ± 2.3	60.1	64.8 ± 3.3	45.5
GeFL-F + HWA + LA	**87.9 ± 1.6**	**81.1**	**76.0 ± 0.7**	**68.8**

Against GeFL-F, the gain is +13.3 (p = 0.004) on MNIST and +17.4 (p = 0.015) on FMNIST. Tail recall rises by +29 and +33. On the paper's own metric (`best_mean_acc`) the gain is +13.5 and +17.3. LA alone accounts for +4.3 and +6.2; HWA adds +9.0 and +11.2 on top of it. Worst-class recall rises from 29.2 / 19.3 (GeFL-F) to 68.9 / 33.5 (MNIST / FMNIST); the Gaussian exact-statistics generator FSG + LA (F01, same seeds) reaches 87.0 / 73.6 balanced accuracy, so on MNIST the repaired CVAE is not clearly better than it. All baselines here are in the GeFL-F family or simple FL references: no dedicated long-tailed FL method and no recalibration of GeFL-F's own heads is compared yet (§B.11, §8).

More clients (F04, MNIST long tail, full schedule, 3 seeds, balanced accuracy):

	K = 10 (F01/F10)	K = 50	K = 100
GeFL-F	74.2	78.1	76.0
+ LA	78.9	81.4	79.5
+ HWA + LA	87.8	86.5	83.1
+ HWA + LA + CSL (ours)	**89.9**	–	83.0 (n = 2)
FSG + LA	87.0	**87.7**	**86.1**
FSG + LA + CSL	–	–	86.3 (n = 1)
MIX + HWA + LA + CSL	–	–	85.3 (n = 2)

The consensus-label arms at K = 50 and 100 were re-run after a bug fix in CSL; the cells marked "–" have no valid post-fix run, and the earlier values are not reported.

- Every variant beats GeFL-F at every K: HWA + LA gains +8.4 at K = 50 and +7.1 at K = 100.
- The best *generator* changes with K. The repaired CVAE-F wins at K = 10. The Gaussian sufficient-statistics generator (FSG) is ahead at K = 50 (87.7 vs 86.5 for HWA + LA) and at K = 100 (86.1 vs 83.1 for HWA + LA and 83.0 for HWA + LA + CSL, 2 seeds).
- On FashionMNIST (K03, 3 seeds) the picture is the same: at K = 50 / 100, HWA + LA scores 75.5 / 73.3, FSG + LA 75.1 / 74.6, GeFL-F 61.9 / 62.0, and adding CSL to HWA + LA *lowers* balanced accuracy (74.5 / 71.2).
- This is the expected structure. FSG's federated estimate equals the centralised one at any K, because it aggregates exact sums. The CVAE's rows, even under HWA, are fitted by

clients that each hold fewer samples as K grows: about 60 images per client at K = 100. The server can see the per-client sample size from aggregated counts, so a rule (CVAE-F when clients are data-rich, FSG when they are data-poor) is possible, but it has not been specified or tested in advance.

SVHN (Kaggle, 3 seeds). Balanced accuracy is 62.45 ± 1.76 for ours, 50.49 ± 2.24 for GeFL-F and 56.44 for + LA, a gain of +12.0. Tail recall is 47.8 against 24.8. On `best_mean_acc` the result is 66.86 vs 57.36 (p = 0.045). The Gaussian generator falls to 44.2. **CIFAR-10** (Kaggle, long tail, 3 seeds; realised imbalance 48:1). GeFL-F collapses hardest here, with a row ratio of 0.06. HWA + LA scores 40.73 against 33.74 for GeFL-F (+7.0) and 37.3 for + LA. The Gaussian generator does better, at 44.17: its exact class means beat a CVAE trained on about 25 tail samples on CIFAR features, the opposite of SVHN. The anchored method is in §B.11.

B.4 GeFL-F's own IID setting (F09)

	Paper GeFL-F	Our GeFL-F	+ CSL	Gain per seed	p
MNIST, K=10	95.47	96.02	**96.71**	+0.57, +0.69, +0.80	0.010
FMNIST, K=10	83.14	82.67	**83.16**	+0.60, +0.47, +0.40	0.016
MNIST, K=50 (seed 0)	95.04	95.16	95.41	+0.26	–
MNIST, K=100 (seed 0)	94.63	94.61	94.72	+0.11	–
SVHN, K=10	76.26	75.82	**76.30**	+0.78, +0.37, +0.29	0.085

Final-round balanced accuracy tells the same story with smaller numbers: MNIST 95.75 → 96.75, FMNIST 82.63 → 83.18 (F10), SVHN 74.54 → 75.15 (K01). Paper values are single published numbers without seeds or error bars.

$\beta = 1$ is worse than $\beta = 0.5$ at every K (96.10 / 95.13 / 94.49), which is the bias term of §5.3. Interleaving synthetic features into real batches (CSLM) is also worse.

On data augmentation. The paper's augmentation results (Table IV: MixUp, CutMix, AugMix, AutoAugment) are for image-space GeFL with DCGAN on CIFAR-10, never for GeFL-F. Augmentation is orthogonal to all three of our changes, so we leave it out of every method.

B.5 One method for both regimes (F10)

Matched seeds and splits, 3 seeds.

	MNIST LT bal.	FMNIST LT bal.	MNIST IID `best_mean_acc`	FMNIST IID `best_mean_acc`
GeFL-F	74.24	58.74	96.01	82.80
+ CSL	76.35	58.59	**96.82**	**83.25**
+ HWA + LA	87.78	75.99	96.01	82.74
+ HWA + LA + CSL (ours)	**89.87**	**76.37**	96.77	83.21
ours − GeFL-F (p)	+15.6 (0.008)	+17.6 (0.019)	+0.76 (0.004)	+0.41 (0.008)

- **Long tail.** The full method beats HWA + LA on every seed: +3.29, +2.21 and +0.77 on MNIST, and +0.28, +0.65 and +0.20 on FMNIST. With 3 seeds this is not yet significant ($p \approx 0.1$). Tail recall is 82.7 against 51.4 on MNIST and 69.8 against 35.6 on FMNIST.
- **CSL alone under the long tail.** It gives +2.1 on MNIST ($p = 0.003$) and nothing on FMNIST. It works best once HWA has made the tail features faithful, as §5.3 predicts: relabelling class-agnostic features cannot help.
- **IID.** The full method is within 0.05 of + CSL. HWA + LA neither helps nor hurts there.

B.6 Regimes, privacy and ablations (K02, Kaggle; minimal variant HWA + LA + CSL)

The gain grows with imbalance (balanced accuracy, K = 10, 3 seeds):

	MNIST: GeFL-F → ours	FashionMNIST: GeFL-F → ours
label skew only (IF = 1, Dir 0.5)	89.8 → 95.6 (+5.8)	74.1 → 79.2 (+5.1)
mild tail (IF = 10)	86.9 → 94.3 (+7.4)	72.6 → 79.4 (+6.8)
heavy tail (IF = 100)	75.3 → 90.2 (+14.8)	58.9 → 76.5 (+17.6)

Dirichlet(0.5) already leaves some clients without some classes, so the non-holder term of Theorem 3 is active even without a global tail.

Ablations (IF = 100):

- *Necessary components.* Removing HWA costs 7–8 points, and removing LA 5–8.
- *HWA's weighting rule matters little.* Effective number, raw counts and uniform weighting over holders give 90.15 / 89.97 / 89.02 on MNIST and 76.51 / 76.45 / 76.33 on FashionMNIST (2 seeds for the alternatives, so these differences are not resolved). This is consistent with the damage coming from *non-holders'* decay.
- *CSL.* It matters where the generator's labels are poor (MNIST: −2.5 without it) and not where they are decent (FashionMNIST).
- *LA temperature.* $\tau = 2$ helps FashionMNIST (+1.4) and slightly hurts MNIST. BBC captures this residual bias without tuning.

Privacy.

- *DP on class counts (minimal variant).* Laplace noise at $\varepsilon = 1$ on every client's class histogram costs 0.2 (FashionMNIST) and 1.3 points (MNIST). At $\varepsilon = 0.1$ the method is still 11–14 points above GeFL-F. This covers the counts HWA uses, not the generator or the heads.
- *Memorisation.* Feature-space MND equals GeFL-F's.

Formal DP on every statistic the anchored method adds (E26; long tail, K = 10, 3 seeds; balanced accuracy).

	MNIST	FashionMNIST
GeFL-F (no formal privacy)	75.25	58.47
final method, exact statistics	94.82	79.67
final method, $(8, 10^{-5})$-DP statistics	90.36	72.77
final method, $(2, 10^{-5})$-DP statistics	85.06	65.59

- *Mechanism.* Features are clipped to the 90th-percentile norm of the public held-out pool. Each of the four releases (counts, sums of h, sums of $\|h\|^2$, sums of $\phi(h)$) gets Gaussian noise, and the budget is split equally under zCDP. Prototypes, MC and KH targets, and the BBC gate are post-processing.

- *Gating noisy classes.* Where a class's noisy statistic would be too noisy to help, MC and KH fall back to the plain generator for that class. This is decided from the noisy counts, at no extra cost. MC is applied to 7 of 10 classes at $\varepsilon = 8$ and to 5 at $\varepsilon = 2$.
- *Cost.* 4.5–6.9 points at $\varepsilon = 8$ and 9.8–14.1 at $\varepsilon = 2$, almost all on the rarest classes, whose 24 samples cannot be both private and accurate. BBC no longer helps under DP.
- *Still ahead, but not everywhere significantly.* At $\varepsilon = 2$ the method is +9.8 above GeFL-F on MNIST and +7.1 on FashionMNIST, where the difference is not significant with 3 seeds ($p = 0.084$).
- *Scope of the guarantee.* The DP covers only the four added statistics, per example, with add/remove neighbours and a trusted party adding the noise. The generator's FedAvg training, the heads, the consensus-label sums and the gate's inputs are not covered, in either method, and secure aggregation is simulated (the server is given the sum).

B.7 What did not work

These ideas were tested on a reduced schedule first and dropped on the evidence. We report them because they sharpen what the working method actually fixes.

- **Prior-completing mixed batches.** These fill a client's missing classes with synthetic features, which works only when tail synthetic features are faithful. Before HWA they are not; FMNIST went from 65.2 to 59.6 with LA.
- **Lazy conditioning decay.** It removes the collapse from the client side, with no shared counts. With HWA present it adds nothing (78.6 vs 79.1). It remains the fallback when even aggregated counts are disallowed.
- **A federated Gaussian generator (exact aggregation of sufficient statistics).** It was competitive on MNIST (87.0 vs 87.9). It is worse in IID data and fails on SVHN ($\approx$ 45 vs 62), where tail fidelity is 0.14: a single Gaussian per class cannot model real feature distributions.
- **A hybrid CVAE/Gaussian generator.** The quick pass favoured it; the full schedule reversed that (87.1 vs 87.9). This shows why a quick pass alone is not enough to decide.
- **Server-side classifier re-calibration, and gap-filling synthetic labels.** Both are weaker than LA.
- **Server-side ensemble distillation on generated features (SED).**
  - *Motivation.* The ensemble of the G heads beats the mean head by 3.2 points on FMNIST IID, 4.5 on MNIST LT and 7.9 on SVHN IID. SED distils that ensemble into every head at the server on fresh generated features, using CSL's target, with no client cost.
  - *Result (FMNIST quick pass).* It *lowers* IID accuracy (81.69 $\rightarrow$ 80.96) and raises the long tail only a little (70.65 $\rightarrow$ 71.30).
  - *Why.* The bound $R_{\text{real}} \leq R_{\text{syn}} + \ell_{\max}\,\mathrm{TV}(p_{\text{syn}}, p_{\text{real}})$ explains it. In IID data the heads are already good, so the generator's infidelity dominates what distillation adds. The ensemble's knowledge can reach the heads only through features as faithful as the generator's.
  - *Implication.* The remaining IID gap to the paper's best FMNIST number (DDPM-F) is a *generator* gap. §B.8 tests exactly that.
- **Bayesian consensus labels.**
  - *Idea.* Replace CSL's fixed mixture by the Bayes posterior under symmetric generator noise, with the generator's fidelity estimated without labels from ensemble agreement.
  - *Result.* Worse than CSL: 83.16 vs 84.22 on FashionMNIST IID ($p = 0.004$), and 87.3 vs 90.5 on MNIST LT.

  - *Why.* The estimate is circular. The ensemble was trained on the generator's own samples, so it agrees with their labels ($\hat{\rho} \to 0.999$), and the target collapses to the hard label.
  - *Lesson.* CSL works as *distillation* of the ensemble's class-similarity structure, not as label correction.

- **Held-class real–synthetic alignment.** Aligning each head's embedding of synthetic features to that of real features on the classes a client holds *increased* the synthetic–real gap on unheld classes. Dropped.

B.8 Generator-agnostic: our method on DDPM-F, the paper's best feature generator

The best FMNIST result of any GeFL variant in the paper is 84.28, from GeFL-F with the feature diffusion model DDPM-F. Our components do not depend on the generator:

- HWA averages the class columns of DDPM-F's two context-embedding layers over holders.
- LA and CSL act on the heads.

We ported the authors' DDPM-F unchanged: the ddpm16 ContextUnet, $n_T = 200$, a linear β schedule from 10^{-4} to 0.02, context dropout 0.1, and Adam at 10^{-4} decayed linearly to 0 over T_{KA} rounds, then frozen. It runs GeFL-F, + CSL and Ours on it. Every round, all clients' synthetic phases draw from one shared pool of fresh uniform-label samples. For each head this is exactly the reference's per-client draw in distribution, at 1/K of the sampling cost, and it is the same for every method compared.

On FMNIST IID (`best_mean_acc`), GeFL-F on DDPM-F scores 83.16 (seed 0 only), not the published 84.28. + CSL scores 83.04 (seed 0) and Ours 82.82 ± 0.39 (seeds 0–1). In our hands DDPM-F is only +0.28 over CVAE-F on the same seed, and CSL does not add to it: diffusion samples are more diverse but less class-pure, with referee fidelity 0.55 against 0.86. On SVHN IID (K04, 3 seeds) Ours on DDPM-F scores 70.37 against 69.77 for GeFL-F on DDPM-F, both well below the CVAE-F pipeline (75.82). So the components *run* on DDPM-F, but we have no evidence that they help it; the section title's "generator-agnostic" means only the former.

B.9 Consensus labels make a larger synthetic budget useful (E16, FMNIST IID, 3 seeds)

Prediction. Train a head on real data plus a share λ of synthetic data. Its error is roughly $\sigma^2/(n_{\text{real}} + \lambda n_{\text{syn}}) + \lambda^2 b(t)^2$, where $b(t)$ is the bias of the synthetic label against the true posterior of a generated feature. The optimal λ grows as b shrinks. Hard labels have a large b, so the paper finds $T_s = 5$ no better than $T_s = 1$ (its Fig. 10). CSL lowers b, so with CSL more synthetic epochs should help.

`best_mean_acc`	$T_s = 1$	$T_s = 3$	$T_s = 5$	$T_s = 10$
GeFL-F (hard labels)	82.96	83.20	83.18	–
+ CSL	83.31	83.90	84.14	**84.37** (84.34 / 84.16 / 84.62)
Ours (HWA + LA + CSL)	83.21	–	84.12	**84.34** (84.55 / 83.88 / 84.59; p = 0.002 vs GeFL-F)

The same budgets under the long tail (FMNIST, IF = 100, Dir 0.5, 3 seeds, final balanced accuracy):

	GeFL-F	Ours, $T_s = 1$	Ours, $T_s = 5$	Ours, $T_s = 10$
Balanced accuracy	59.13	76.94	78.76	**79.23**
Tail recall	36.04	70.22	71.92	71.83
Worst class	20.36	29.38	32.11	32.79

Under the long tail the larger budget also helps: +2.29 over $T_s = 1$, and +20.1 over GeFL-F. On MNIST under the long tail it does not (90.20 / 90.78 / 90.15). There, HWA has made the tail features faithful and CSL has lowered their label bias. This is the synergy of §5.3, seen through the budget.

- The interaction appears exactly as predicted.
- At $T_s = 10$, CSL with the cheap CVAE-F generator reaches the paper's best FMNIST number of *any* variant, 84.28 (GeFL-F with feature diffusion), on `best_mean_acc` (final round: 84.32). We have no equivalence test, so this is a numerical match against a single published value, not a demonstrated tie.
- MNIST is near its ceiling (oracle ≈ 97.2), and a larger budget does not help there.
- With one fixed setting ($T_s = 5$), the full method gives 84.12 on FMNIST and 96.6 on MNIST. With $T_s = 10$ the full method gives 84.34 on FMNIST, a tie with the paper's best (84.28).

B.10 Against every GeFL variant in the paper (IID, `best_mean_acc`)

The paper's Figure 4 has ten methods: FedProx, FedALA, and GeFL and GeFL-F each with DCGAN, CVAE, DDPM w = 0 and DDPM w = 2. In each setting, the comparison is with the best of them.

Setting	GeFL-F CVAE-F	Best in paper (which)	Ours	Ours − best
MNIST, K=10	95.47	96.44 (GeFL, image DDPM)	**96.77**	+0.33
MNIST, K=50	95.04	95.04 (GeFL-F CVAE-F)	**95.41** (CSL, 1 seed)	+0.37
MNIST, K=100	94.63	94.63 (GeFL-F CVAE-F)	94.72 (CSL, 1 seed)	+0.09
FMNIST, K=10	83.14	84.28 (GeFL-F DDPM-F)	**84.34** (Ours, $T_s = 10$); 84.12 ($T_s = 5$)	+0.06 / −0.16 (tie)
SVHN, K=10	76.26	76.26 (GeFL-F CVAE-F)	76.30 (CSL)	+0.04 (tie)
CIFAR-10, K=10	55.86	59.36 (GeFL, image DDPM); 62.67 with MixUp	60.93 (Ours-A, $T_s = 10$)	+1.57 / −1.74

The paper's numbers are single published values; ours are 3-seed means of the best round. At the final round the gaps shrink: on CIFAR-10, Ours-A at $T_s = 10$ ends at 56.82 against 56.68 for our GeFL-F (+0.14); Ours-A at $T_s = 1$ ends at 58.62. On MNIST, Ours-A ends at 97.58 against 95.75.

The two strongest CIFAR-10 numbers come from **image-space** GeFL, with an image diffusion model or with DCGAN + MixUp. Neither has a shared feature extractor, and the paper notes that GeFL-F on CIFAR-10 is limited by its small FE. Our method keeps GeFL-F's FE and privacy model. The fair claims are therefore:

1. over GeFL-F with the same FE and generator;
2. over the best of *all* variants wherever the FE is not the bottleneck: MNIST, SVHN, and FMNIST pending DDPM-F.

B.11 The anchored method (E20, E24, E25; K = 10, 3 seeds unless marked)

	MNIST LT	FMNIST LT	MNIST IID	FMNIST IID
GeFL-F	75.25	58.47	96.01	82.76
minimal variant, CVAE + HWA + LA + CSL	90.40	76.60	96.77	83.19
minimal variant, $T_s = 10$	90.15	79.23	–	**84.34**
PC-VAE + MC	92.31	77.56	–	82.92
KME-Gen, server-only generator (1 seed)	89.88	75.11	–	81.82
Ours-A: PC + MC + KH + LA + CSL, $T_s = 1$	93.11	78.22	96.67	83.35
Ours-A, $T_s = 10$	**94.82**	**79.67**	**97.62**	83.96
Ours-A, $T_s = 10$**, + gated BBC**	**95.17**	**80.55**	(BBC off)	(BBC off)
oracle: its heads, last layer re-fit on pooled real data	94.93	81.55	97.78	84.09

Long-tail columns report final balanced accuracy and IID columns report `best_mean_acc` (the paper's metric; final round: MNIST 97.58 for Ours-A at $T_s = 10$ against 95.75 for GeFL-F). The MNIST IID GeFL-F and minimal-variant numbers come from F10 (same seeds). Worst-class recall under the long tail: Ours-A $T_s = 10$ 87.1 / 32.8, GeFL-F 26.8 / 18.6 (MNIST / FMNIST). The oracle row re-fits the last layer of *our* heads; the same oracle applied to plain GeFL-F is the next paragraph.

Against a recalibrated GeFL-F (the decisive comparison). Re-fitting only the last layer of *plain GeFL-F's* heads on pooled real embeddings (the same oracle) gives 90.7 / 79.2 / 64.8 balanced accuracy on MNIST / FMNIST / SVHN under the long tail (GeFL-F reference runs F10 and K01, same splits and seeds). Ours-A's margin over that refit is +4.1 ($p = 0.01$), +0.5 ($p = 0.35$) and +0.8 ($p = 0.29$; SVHN with the rows branch), and oracle against oracle it is +4.2 / +2.4 / +3.6. Against the other GeFL-F replicate (E20) the first two margins are +3.6 ($p = 0.03$) and +0.7 ($p = 0.23$). Most of GeFL-F's long-tail loss is therefore a *readout* failure that a last-layer recalibration largely repairs. The oracle uses pooled real data and is not federated; an exact federated readout (ER, an LDA readout from secure-aggregated embedding sums) came within 0.8 of it in a one-seed, reduced-schedule pilot (T03; `docs/research/readout_report.md`). Until a federated recalibration baseline is run at full schedule (job B1 of the action plan), the gains over GeFL-F above should be read as gains over an un-recalibrated baseline.

Against GeFL-F and the minimal variant.

- Over GeFL-F: +19.6 ($p = 0.002$) and +21.2 ($p = 0.011$) under the long tail. In IID: +1.6 on MNIST ($p = 0.001$; 97.62, +1.18 over the best of all ten methods in the paper) and +1.2 on FashionMNIST ($p = 0.021$).
- Over the minimal variant: +4.7 and +0.5 under the long tail (positive on every seed). In IID: +0.85 on MNIST ($p = 0.001$) and −0.4 on FashionMNIST ($p = 0.18$).

- *Why the IID gap.* In balanced data FedAvg does not collapse class rows, so the CVAE's class-specific parameters cost nothing and add detail.

Mechanism, measured.

- *Raw samples.* The anchored decoder's raw samples are under-dispersed (tail spread 0.39–0.49 of real). On FashionMNIST they are also *purer than real data*: referee fidelity 0.89–0.91, against the referee's 0.84 accuracy on real test data.
- *After MC and KH.* Spread is restored to 0.93–0.98, and fidelity moves to real-data difficulty (IID: 0.837 vs 0.839).
- *KH.* In the component ablation (K09, 3 seeds) removing KH lowers balanced accuracy by 0.71 on MNIST ($p = 0.059$) and 0.30 on FashionMNIST ($p = 0.097$): a small effect, not significant. Across all five removals in K09 the cluster-level sign test gives $p = 0.031$ that removing a component hurts (the earlier "30/30, $p = 2 \cdot 10^{-9}$" treated dependent comparisons as independent).
- *Proposition 11.* With the anchored generator, $T_s = 10$ adds +1.70 on MNIST LT ($p = 0.04$), where it added −0.05 to the CVAE (tail fidelity 0.28). The interaction is +1.75, positive on every seed. The same holds on MNIST IID: +0.95 for the anchored generator, against −0.18 for consensus labels on the CVAE.

More clients (E22, MNIST long tail, K = 100, 3 seeds).

	GeFL-F	CVAE + HWA + LA	FSG + LA	Ours-A, $T_s = 1$	**Ours-A, $T_s = 10$**
balanced accuracy	75.98	83.07	86.05	90.41	**94.30**
tail recall	50.3	66.6	76.0	83.4	**90.5**

- *Robust to K.* Ours-A loses only 0.5 points from K = 10 to K = 100, while the CVAE + HWA loses 4.7.
- *Margins.* +18.3 over GeFL-F ($p = 0.0004$) and +8.3 over the Gaussian generator ($p = 0.001$).
- *Proposition 11 again.* The synthetic budget is worth +3.9 at K = 100 against +1.7 at K = 10: the optimal synthetic share $w^\star \propto 1/n_{\text{real}}$ grows as each client holds fewer real samples (about 60 here).

The hybrid generator where the anchor works (E29, same seeds as E25 / E22). PCR adds HWA-aggregated class rows to the anchored decoder. Paired against Ours-A at $T_s = 10$, it changes nothing measurable:

	Ours-A	hybrid PCR	difference
MNIST LT, K = 10	94.82	95.03 (BBC 95.34)	+0.22 ($p = 0.29$)
MNIST LT, K = 100 (2 seeds)	94.25	94.36	+0.10
FashionMNIST LT	79.67	79.60 (BBC 80.46)	−0.07 ($p = 0.51$)
FashionMNIST IID (`best_mean_acc`)	83.96	84.11	+0.15 ($p = 0.28$)

- *No cost.* The rows cost nothing where the anchor is informative. At K = 100 they do not bring back the CVAE's degradation with the number of clients: HWA keeps them from collapsing, and the anchor carries the location.

- *Feature space.* The shared FE's feature space confirms the pixel-space ordering of class-mean separation. F is 0.29 (MNIST) and 0.53–0.60 (FashionMNIST), and nearest-class-mean accuracy is 81–82% and 68–69%. The E30 smoke run gives $F = 0.025$ and 11.5% on SVHN.
- *What it means.* If PCR or MC-S also closes the SVHN gap (Kaggle K10, K11), a single generator serves every dataset.

CIFAR-10 and SVHN (Kaggle K01, K08; 3 seeds, same seeds and splits).

	CIFAR-10 LT	SVHN LT	CIFAR-10 IID	SVHN IID
GeFL-F	33.74	50.49	59.10	75.82
+ HWA + LA	40.73	**62.45**	–	75.64
minimal variant (CVAE + HWA + LA + CSL) / + CSL	42.10	–	59.85	**76.30**
FSG + LA (Gaussian, exact statistics)	44.17	44.23	–	–
Ours-A, $T_s = 1$	**46.06**	55.89	60.39	73.86
Ours-A, $T_s = 1$, + gated BBC	**48.71**	57.19	(off)	(off)
Ours-A, $T_s = 10$	44.38	54.41	**60.93**	74.38
Ours-A, $T_s = 10$, + gated BBC	47.26	54.86	(off)	(off)
paper: GeFL-F / best of all ten methods	–	–	55.86 / 59.36	76.26 / 76.26

Long-tail columns report final balanced accuracy and IID columns report `best_mean_acc` (final round, CIFAR-10 IID: GeFL-F 56.68, Ours-A 58.62 at $T_s = 1$ and 56.82 at $T_s = 10$). The per-seed GeFL-F records for K01 CIFAR-10 were not copied back, so tests against them are unpaired (Welch, from means and standard deviations); all other tests are paired.

- *CIFAR-10: the anchored method wins in both regimes.* Under the long tail it is +12.3 over GeFL-F ($p = 0.001$), +4.0 over the minimal variant ($p = 0.017$) and +1.9 over FSG ($p = 0.11$). With BBC it is +4.54 over FSG. Its tail fidelity after KH is 41%, against 23% for the CVAE + HWA. In IID it reaches 60.93 on the best round, but only +0.14 over our GeFL-F at the final round, so we claim no IID gain on CIFAR-10.
- *A registered prediction that failed.* Before these runs we predicted a tie or a small loss on CIFAR-10, because its class means are weak in pixel space (nearest-class-mean accuracy 27%). It won. What decides is the anchored generator's fidelity *relative to the CVAE's*, and on CIFAR-10 the CVAE's own fidelity is low.
- *SVHN: the anchored generator fails.* It is −6.6 against + HWA + LA under the long tail and −2.0 against GeFL-F in IID. This is Proposition 12: SVHN's anchors are $\sqrt{2F} \approx 0.1$–0.16 within-class spreads apart, and the synthetic classes are nearly indistinguishable (tail fidelity 24%).
- *The synthetic budget on hard data.* On CIFAR-10 and SVHN, $T_s = 10$ raises the best-round metric in all four settings but lowers final balanced accuracy (CIFAR-10 long tail: 44.4 against 46.1, $p = 0.03$). Proposition 11 predicts this: $w^\star = \sigma^2/(2n_r b^2)$ falls with the generator's error b, and there tail fidelity after KH (24–46%) is far below real-data accuracy. On MNIST and FashionMNIST, where fidelity matches real data, $T_s = 10$ is better on both metrics.

- *Not beaten.* Image-space GeFL + MixUp (62.67 on CIFAR-10 IID, Table IV) remains above every result here. It uses a different pipeline (no shared feature extractor) and data augmentation, which is orthogonal to our method.

SVHN: MC-S, the hybrid and the gate (Kaggle K10, K11; 3 seeds, same seeds and splits).

SVHN, K = 10	long tail	+ gated BBC	IID (`best_mean_acc`)
GeFL-F	50.49	–	75.82
+ HWA + LA	62.45	–	75.64
Ours-A (PC + MC + KH)	55.76	57.27	74.38
Ours-A-S (PC + **MC-S** + KH)	61.80	63.30	77.05
hybrid PCR + MC + KH	60.86	62.08	76.06
hybrid PCR + MC-S + KH	62.89	64.57	**77.39**
rows (CVAE + HWA) + MC + KH, $T_s = 1$: the final method at $F < 0.05$	**65.60**	**66.96**	77.07 (final 75.89)
paper: GeFL-F / best of all ten methods	–	–	76.26 / 76.26

- *MC-S works where Proposition 12 says the mean cannot.* Against Ours-A it adds +6.0 under the long tail (every seed, +4.5 to +7.0, $p = 0.016$) and +2.7 in IID ($p = 0.026$). The proxy and the QDA measurement predicted the direction.
- *Rows are better still where means are uninformative.* The CVAE + HWA with MC + KH gives 65.60, ahead of the hybrid with MC-S at 62.89 (Welch $p = 0.07$). The hybrid's anchor leaves its raw samples under-dispersed. On SVHN, MC + KH then lowers its tail fidelity (35 → 32%), while they raise the CVAE's (36 → 42%).
- *IID.* 77.07 for the final method and 77.39 for the best variant on the best round, against a single published 76.26; at the final round the final method scores 75.89 against 74.54 for our GeFL-F.
- *The gate.* Every SVHN run reports F between 0.005 and 0.013 and nearest-class-mean accuracy 12%. The F-gate therefore selects the rows on SVHN and the anchor on MNIST and FashionMNIST, which is the better branch wherever both were measured. On CIFAR-10 ($F = 0.055$–0.078, close to the threshold) only the anchor branch has been run.
- *BBC values* in this section are 3-seed means from the per-seed records.

VCT and MixUp (K13, CIFAR-10, 3 seeds). Neither helps. Under the long tail Ours-A + MixUp scores 46.20 and Ours-A + VCT 45.65 (Ours-A alone: 46.06). In IID, final-round balanced accuracy falls to 54.76 (MixUp) and 53.18 (VCT), against 56.82 for Ours-A at $T_s = 10$. VCT is therefore not part of the method.

Pending.

- A federated recalibration baseline for GeFL-F (B1 in `docs/review/action_plan.md`; needs re-runs, since no models were saved before the A12 code change).
- K15: optimiser-level fixes of the collapse (decoupled decay, no decay) against HWA.
- K14: component ablation with more seeds and the covariance correction on the class rows.
- K12: the gated method, run end to end on all four datasets.
- External long-tailed FL baselines on the same frozen extractor (FedLC / FedRoD-style losses, FedRS, CCVR, FedProto).

B.12 Cost

- *Minimal variant.* Summed over all stages, HWA + LA + CSL takes 373 s per run against GeFL-F's 366 s (+1.9%), measured on MNIST under the long tail, averaged over 3 seeds. HWA adds C numbers per client per round; CSL adds the per-round head or pool-prediction upload of §5.3 and one broadcast of soft labels.
- *Ours-A.* End to end it is 2.2–2.8× slower than GeFL-F. The head stage (Stage III) is 5–14× slower, and kernel herding accounts for 83–88% of it.
- *Bytes.* The PC-VAE has 6.93M parameters on CIFAR-10 against 5.61M for the CVAE-F (+23% per generator upload). The exact statistics (counts, sums of h, of $\|h\|^2$ and of $\phi(h)$) take 0.28–0.35 MB per client in two secure-aggregation rounds. MC-S adds a broadcast $D \times D$ pooled covariance (5.0–28.8 MB).

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